বহু গুতাগুতির কারণে এই নোট করসি।
সো ভুলভাল থাকলে আমার দোষ নাই।
GPT, Claude, Deepseek, বই, slide, classnote সবকিছুর জগাখিচুরি।
কিছু জিনিস এক্সট্রা আছে, কিছু জিনিস নাই।
Frequency Analysis এর slide এর ... লাস্ট কয়েক পেইজ নাই।
নিজ দায়িত্বে যা যা দরকার যেমনে দরকার মিলায় নিয়েন, mujhse aur nehi hota

Frequency Domain Representation of LTI DTS

For an LTI system, $y[n] = x[n] * h[n]$ → time domain analysis tells how the samples combine.

Frequency domain analysis tells how each frequency component is modified by the system.

→A sinusoidal signal can be expressed in terms of an exponential signal.

▶ **Eigenfunction**

An input $x[n]$ is an eigenfunction of a system if the output has the same form as input, only multiplied by a constant.

$$T\{x[n]\} = \lambda\, x[n]$$

ex- If input is $e^{j\omega N}$, the output = $c \times e^{j\omega N}$ where c is independent of n.

Proving the frequency response is the eigenvalue and $e^{j\omega N}$ is eigenfunction

For an LTI system,

$$y[n] = \sum_{K=-\infty}^{\infty} h[K]\, x[n-K]$$

Let, $x[n] = e^{j\omega n}, \quad -\infty < n < \infty$

$$\therefore y[n] = \sum_{K=-\infty}^{\infty} h[K]\, e^{j\omega(n-K)} = \sum_{K=-\infty}^{\infty} h[K]\, e^{j\omega n} \cdot e^{-j\omega K}$$

$$= e^{j\omega n} \sum_{K=-\infty}^{\infty} h[K]\, e^{-j\omega K} \qquad [\because e^{j\omega n} \text{ not dependent on sum variable } k]$$

Here, $\sum_{K=-\infty}^{\infty} h[K]\, e^{-j\omega K}$ depends on ω not on n, thus quantity is called the frequency response.

$\therefore$ Frequency Response, $H(e^{j\omega}) = \sum_{K=-\infty}^{\infty} h[K] \cdot e^{-j\omega K}$ → DTFT or fourier transform of impulse response $h[n]$

Substituting,

$$y[n] = \underbrace{H(e^{j\omega})}_{\text{eigenvalue}}\ e^{j\omega n}$$ ← eigenfunction of LTI

Input $x[n] = e^{j\omega n}$

Output $y[n] = \underbrace{H(e^{j\omega})}_{\text{constant}} \cdot e^{j\omega n}$

▷ What does $H(e^{j\omega})$ do?

$H(e^{j\omega}) = 2e^{-j\pi/4}$

Input, $x[n] = e^{j\omega n}$

$\therefore y[n] = 2e^{-j\pi/4} \cdot e^{j\omega n}$

$= 2e^{j(\omega n - \frac{\pi}{4})}$

- 2 → amplitude changes from 1 to 2
- $-\frac{\pi}{4}$ → Phase becomes $0 \to -\frac{\pi}{4}$

frequency, ω remains unchanged

Magnitude phase form

Any complex number can be written as-

$$H(e^{j\omega}) = H_{re}(e^{j\omega}) + H_{im}(e^{j\omega})$$

$$= \underbrace{|H(e^{j\omega})|}_{\text{magnitude response}} \cdot e^{\underbrace{j\theta(\omega)}_{\text{phase response}}}$$

where,

$|H(e^{j\omega})|$ = amplitude gain

$\theta(\omega)$ = phase shift

Since H is complex, it can be written as

$$H = H_R + jH_J$$

$\therefore$ Magnitude $= \sqrt{H_R^2 + H_J^2}$

- $|H(e^{j\omega})| = \sqrt{H_R^2(e^{j\omega}) + H_J^2(e^{j\omega})}$ called the magnitude response.
- $\theta(\omega) = \tan^{-1}\left(\frac{H_I}{H_R}\right)$ which is the phase response

$\therefore$ $H(e^{j\omega}) = |H(e^{j\omega})| \cdot e^{j\theta(\omega)}$ which is the polar form.

Gain, $g(\omega) = 20\log_{10}|H(e^{j\omega})|$ dB
is the magnitude response in decibels.

- If $|H| > 1$, $g(\omega) > 0$ → gain/amplification
- If $|H| = 1$, $g(\omega) = 0$ → no magnitude change
- If $|H| < 1$, $g(\omega) < 0$ → attenuation/loss

If $h[n]$ is real, then, $H(e^{-j\omega}) = H^*(e^{j\omega})$

- Magnitude response is even, $|H(e^{j\omega})| = |H(e^{-j\omega})|$
- Phase response is odd, $\theta(\omega) = -\theta(-\omega)$

In an LTI system,

$$y[n] = h[n] \circledast x[n]$$

Taking DTFT on both sides,

$$Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})$$

and, $H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$

DTFT → how much of each frequency is inside this signal

DTFT of $x[n] \Rightarrow X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$

$\therefore$ Convolution in time-domain = Multiplication in frequency domain.

1.54

① Prove that, $u[n] = z^n$ is an eigenfunction if z is a complex constant.

Given input, $u[n] = z^n$

the LTI output is, $y[n] = \sum_k h[k]\, u[n-k]$

$$= \sum_k h[k]\, z^{n-k} = \sum_k h[k]\, z^n \cdot z^{-k}$$

$$= z^n \sum_k h[k] \cdot z^{-k}$$

The summation is independent of n. So, if $H(z) = \sum_k h[k] z^{-k}$

$$y[n] = H(z) \cdot z^n$$

which is why, z^n is an eigenfunction.

② If $v[n] = z^n \mu[n]$, is it an eigenfunction?

Given, $v[n] = z^n \mu[n]$

∴ The LTI output is $y[n] = \sum_k h[k]\, v[n-k]$

$$= \sum_k h[k] \cdot z^{n-k} \cdot \mu[n-k]$$

$$= z^n \sum_k h[k] z^{-k} \cdot \mu[n-k]$$

Here, $\mu[n-k]$ is dependent on n and $\sum_k h[k] z^{-k} \mu[n-k]$ is not a constant independent of n.

So, we cannot write it in terms of constant x input.

The $v[n]$ is not an eigenfunction.

4.60 | (a) Given,

$$x[n] = \cos(\omega_0 n)$$

$$y_a[n] = \cos\left(\frac{\pi n}{5}\right) x[n]$$

$$= \cos\left(\frac{\pi n}{5}\right) \cos\omega_0 n$$

$$= \frac{1}{2}\left[\cos\left(\frac{\pi n}{5} - \omega_0 n\right) + \cos\left(\frac{\pi n}{5} + \omega_0 n\right)\right]$$

$$= \frac{1}{2}\left[\cos\left[\left(\omega_0 - \frac{\pi}{5}\right)n\right] + \cos\left[\left(\omega_0 + \frac{\pi}{5}\right)n\right]\right]$$

So the frequencies are $\left(\omega_0 \pm \frac{\pi}{5}\right)$

(b) Given,

$$y_b[n] = x^4[n]$$

$$= \cos^4(\omega_0 n) = (\cos^2(\omega_0 n))^2$$

$$= \left\{\frac{1}{2}(1+\cos(2\omega_0 n))\right\}^2$$

$$= \frac{1}{4}\left[1 + 2\cos(2\omega_0 n) + \cos^2(2\omega_0 n)\right]$$

$$= \frac{1}{4}\left[1 + 2\cos(2\omega_0 n) + \frac{1}{2}(1+\cos(4\omega_0 n))\right]$$

$$= \frac{1}{4}\left[1 + 2\cos(2\omega_0 n) + \frac{1}{2} + \frac{1}{2}\cos(4\omega_0 n)\right]$$

$$= \frac{3}{8} + \frac{1}{2}\cos(2\omega_0 n) + \frac{1}{8}\cos(4\omega_0 n)$$

∴ The frequencies present are $2\omega_0$, $4\omega_0$ and 0 [for the constant].

(c) Given,

$$y_c[n] = x[4n]$$

$$= \cos(4\omega_0 n)$$

∴ The frequency is $4\omega_0$

Frequency Response of LTI - DTS (FIR)

We know, a finite impulse response,

$$y[n] = \sum_{K=N_1}^{N_2} h[K]\, x[n-k] \quad \text{where } N_1 < N_2$$

Taking fourier transform on both sides,

$$Y(e^{j\omega}) = \underbrace{\sum_{K=N_1}^{N_2} h[K]\, e^{-j\omega k}}_{H(e^{j\omega})} \cdot X(e^{j\omega})$$

$Y(e^{j\omega}) = \mathcal{F}\{y[n]\}$
$X(e^{j\omega}) = \mathcal{F}\{x[n]\}$
$H(e^{j\omega}) = \mathcal{F}\{h[n]\}$

$$\therefore Y(e^{j\omega}) = H(e^{j\omega}) \cdot X(e^{j\omega})$$

Here, $H(e^{j\omega}) = \sum_{K=N_1}^{N_2} h[n]\, e^{-j\omega K}$ which is defined for $N_1 < K < N_2$

4.56

① $h[n] = \delta[n] - \alpha\,\delta[n-R]$

Determine the frequency response $G_1(e^{j\omega})$ where $g[n] = h[n] \circledast h[n] \circledast h[n]$

We know, convolution in time domain = multiplication in frequency domain.

$$\therefore g[n] = h[n] \circledast h[n] \circledast h[n]$$

is equivalent to

$$G_1[e^{j\omega}] = H(e^{j\omega}) \cdot H(e^{j\omega}) \cdot H(e^{j\omega})$$

Now, to find $H(e^{j\omega})$, we use fourier transform of $h[n]$

$$\therefore H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]\, e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} (\delta[n] - \alpha\,\delta[n-R]) e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} \delta[n] \cdot e^{j\omega n} - \alpha \sum_{n=-\infty}^{\infty} \delta[n-R]\, e^{j\omega n}$$

$$= e^{-j\omega(0)} - \alpha e^{-j\omega R} \quad [\because \delta[n]=1 \text{ only at } n=0$$
$$= 1 - \alpha e^{-j\omega R} \quad \delta[n-R]=1 \text{ at } n=R]$$

$$\therefore H(e^{j\omega}) = 1 - \alpha e^{-j\omega R}$$

$$\therefore G_1(e^{j\omega}) = H(e^{j\omega}) \cdot H(e^{j\omega}) \cdot H(e^{j\omega})$$
$$= (1 - \alpha e^{-j\omega R})^3$$
$$= 1 - 3e^{-j\omega R} + 3e^{-2j\omega R} + e^{-3j\omega R}$$

(ANS)

Shortcuts worth remembering :

$$\delta[n] \longleftrightarrow 1$$
$$\delta[n-k] \longleftrightarrow e^{-j\omega k}$$
$$a^n \mu[n] \longleftrightarrow \frac{1}{1-ae^{-j\omega}} \quad |a|<1$$
$$cx[n] \longleftrightarrow cX(e^{j\omega})$$
$$x[n-n_0] \longleftrightarrow e^{-j\omega n_0} \cdot X(e^{j\omega})$$

4.64

② Consider the cascade of two causal LTI systems

$$h_1[n] = \alpha\delta[n] + \delta[n-1]$$
$$h_2[n] = \beta^n \mu[n] \quad \text{where } |\beta|<1$$

Determine the frequency response $H(e^{j\omega})$ of the overall system. For what values α and β, will $|H(e^{j\omega})| = 1$?

Given,

$$h_1[n] = \alpha\delta[n] + \delta[n-1]$$
$$h_2[n] = \beta^n \mu[n] \qquad |\beta|<1$$

$$\therefore H_1(e^{j\omega}) = \alpha \cdot 1 + e^{-j\omega} \cdot 1 = \alpha + e^{-j\omega}$$

$$H_2(e^{j\omega}) = \sum_{K=0}^{\infty} h_2[n]\, e^{-j\omega k}$$

$$= \sum_{K=0}^{\infty} \beta^K \mu[K] \cdot e^{-j\omega k}$$

$$= \sum_{K=0}^{\infty} (\beta e^{-j\omega})^K$$

$$= \frac{1}{1-\beta e^{-j\omega}}$$

$$\therefore H(e^{j\omega}) = H_1(e^{j\omega}) \cdot H_2(e^{j\omega}) = \frac{\alpha + e^{-j\omega}}{1-\beta e^{-j\omega}}$$

According to question,

$$\left| \frac{\alpha + e^{-j\omega}}{1-\beta e^{-j\omega}} \right| = 1$$

or, $$\frac{|\alpha + e^{-j\omega}|}{|1-\beta e^{-j\omega}|} = 1$$

or, $|\alpha + e^{-j\omega}| = |1-\beta e^{-j\omega}|$

or, $|\alpha + e^{-j\omega}|^2 = |1-\beta e^{-j\omega}|^2$

or, $(\alpha + e^{-j\omega})(\alpha + e^{+j\omega}) = (1-\beta e^{-j\omega})(1-\beta e^{j\omega})$ $\quad [\because |z|^2 = z \cdot z^*]$

or, $\alpha^2 + \alpha e^{j\omega} + \alpha e^{-j\omega} + e^{-j\omega+j\omega} = 1 - \beta e^{+j\omega} - \beta e^{-j\omega} + \beta^2 e^{-j\omega+j\omega}$

or, $\alpha^2 + \alpha(e^{j\omega} + e^{-j\omega}) + 1 = 1 - \beta(e^{j\omega} + e^{-j\omega}) + \beta^2$

or, $\alpha^2 + 2\alpha\cos\omega = \beta^2 - 2\beta\cos\omega$ $\quad [\because e^{-j\omega} + e^{j\omega} = 2\cos\omega]$

or,

For this to be true, for every ω, the constants and co-efficients must match

Equating both sides.

i. $\alpha^2 = \beta^2$

ii. $2\alpha = -2\beta$

or, $\alpha = -\beta$ which satisfies ①

∴ Required condition, $\alpha = -\beta$ and $|\beta| < 1$

Frequency Response of the moving average filter:

Moving average filter is written as:

$$y[n] = \frac{1}{M} \sum_{l=0}^{M-1} x[n-l]$$

To find the impulse response $h[n]$ of the moving average filter.

$$y[n] = \frac{1}{m} [x[n] + x[n-1] + \cdots + x[n-(m-1)]$$

and the general FIR form, we can write this as,

$$y[n] = h[0]x[n] + h[1]x[n-1] + \cdots + h[m-1]x[n-(m-1)]$$

i.e. $h[0] = h[1] = \cdots = h[m-1] = \frac{1}{m}$

and outside that interval, $h[n] = 0$

$$\therefore h[n] = \begin{cases} \frac{1}{M} & 0 \le n \le m-1 \\ 0 & \text{otherwise} \end{cases}$$

∴ The moving average filter is FIR.

Now,

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n}$$

∴ The frequency response for Moving Average Filter,

$$H(e^{j\omega}) = \sum_{n=0}^{m-1} h[n]\, e^{-j\omega n} \quad [\because h[0] = 0 \text{ except } 0 \le n \le m-1]$$

$$= \sum_{n=0}^{m-1} \frac{1}{m} e^{-j\omega n}$$

$$= \frac{1}{m} \sum_{n=0}^{m-1} e^{-j\omega n} = \frac{1 - e^{-j\omega m}}{m(1 - e^{-j\omega})} \quad \left[\because \sum_{n=0}^{m-1} r^n = \frac{1-r^m}{1-r}\right]$$

Factor $e^{-j\omega m/2}$ from the numerator and $e^{-j\omega/2}$ from the denominator,

$$H(e^{j\omega}) = \frac{1}{m} \frac{e^{-\frac{j\omega m}{2}}\left(e^{+\frac{j\omega m}{2}} - e^{-j\omega m/2}\right)}{e^{-j\omega/2}\left(e^{+j\omega/2} - e^{-j\omega/2}\right)}$$

$$= \frac{1}{m} \frac{e^{-j\omega m/2}\left(2j\sin\left(\frac{m\omega}{2}\right)\right)}{e^{-j\omega/2}\left(2j\sin\left(\frac{\omega}{2}\right)\right)} \quad \left[\because e^{j\theta} - e^{-j\theta} = 2j\sin\theta\right]$$

$$= \frac{1}{m} e^{-j(m-1)\omega/2} \cdot \frac{\sin\left(\frac{m\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}$$

∴ Magnitude response, $|H(e^{j\omega})| = \left|\frac{1}{m} \frac{\sin\left(\frac{m\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}\right|$

∴ Phase response, $\theta(\omega) = -\frac{(m-1)\omega}{2} + \pi \sum_{K=1}^{\lfloor m-2 \rfloor} \mu\left(\omega - \frac{2\pi k}{m}\right)$.

where $\mu\left(\omega - \frac{2\pi k}{m}\right) = \begin{cases} 1 & \omega \ge \frac{2\pi k}{m} \\ 0 & \text{otherwise} \end{cases}$

why?

$H(e^{j\omega}) = \underbrace{\frac{1}{m} \frac{\sin\left(\frac{m\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}}_{\text{real valued}} \cdot \underbrace{e^{-j\omega(m-1)/2}}_{\text{complex}}$

real valued part can be
- +ve → phase 0
- -ve → phase π

ex- if real valued part = -2
we write $2e^{j\pi}$
and $\sin\left(\frac{\omega m}{2}\right)$ becomes 0 when $\frac{m\omega}{2} = K\pi$
$\therefore \omega = \frac{2\pi k}{m}$

Steady-State and Transient Response

A total response maybe written as,

$$y[n] = y_{ss}[n] + y_{tr}[n]$$

where, $y_{ss}[n]$ = steady state / long-term response

$y_{tr}[n]$ = transient / temporary response

- Transient response is the temporary beginning behaviour right after the input starts.
- steady state response is the behavior that remains after the temporary part dies out.

→ For a stable system, the transient part eventually dies out i.e $y_{tr} \to 0$ and $y[n] \approx y_{ss}[n]$

① $h[n] = \{4, -5, 6, -3\}$ $0 \le n \le 3$

$$\therefore y[n] = h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + h[3]x[n-3]$$

$$= 4x[n] - 5x[n-1] + 6x[n-2] - 3x[n-3]$$

At $n=0$,

$$y[0] = 4x[0] - 5x[-1] + 6x[-2] - 3x[-3] = 4 \quad [\because x[n] = \mu[n]]$$

Similarly,

$$y[1] = 4\mu[1] - 5\mu[0] + 6\mu[-1] - 3\mu[-2] = 4 - 5 = -1$$

$$y[2] = 4\mu[2] - 5\mu[1] + 6\mu[0] - 3\mu[-1] = 4 - 5 + 6 = 5$$

$$y[3] = 4\mu[3] - 5\mu[2] + 6\mu[1] - 3\mu[0] = 4 - 5 + 6 - 3 = 2$$

and, for $n \geq 3$, all 4 step terms remain equal to 1

$\therefore y[n] = 2 \quad n \geq 3$

$\therefore y[n] = 4, -1, 5$ belong to transient behavior

$H(e^{j0}) = \sum_n h[n]$

$H(e^{j0}) = 2$

$\omega = 0 \rightsquigarrow$ behaves like a DC signal

▷ Develop an expression for the steady state response of an LTI system DTS with a real impulse response $h[n]$ for a sinusoidal input in terms of frequency response function, $H(e^{j\omega})$.

Suppose,

$$x[n] = A\cos(\omega_0 n + \phi)$$

A = input amplitude
ω_0 = input frequency
ϕ = input phase

For an LTI system with frequency response, $H(e^{j\omega})$,

$$H(e^{j\omega_0}) = |H(e^{j\omega_0})|\, e^{j\theta(\omega_0)}$$

Here, $x[n] = A\cos(\omega_0 n + \phi)$

$$= \frac{1}{2}\left(A e^{j(\omega_0 n+\phi)} + e^{-j(\omega_0 n+\phi)}\right) \quad [\because \cos\theta = \tfrac{1}{2}(e^{j\theta} + e^{-j\theta})]$$

Now, as $e^{j\omega n}$ is an eigenfunction, we can write,

$$y[n] = \frac{A}{2}\left(e^{j\phi}\cdot H(e^{j\omega_0})\, e^{j\omega_0 n} + e^{-j\phi}\, H(e^{j\omega_0})\cdot e^{-j\omega_0 n}\right)$$

For real $h[n]$,

$$H(e^{-j\omega_0}) = H^*(e^{j\omega_0})$$

let, $H(e^{j\omega_0}) = |H|e^{j\theta}$

$\therefore H(e^{-j\omega_0}) = |H|e^{-j\theta}$

$$\therefore y[n] = \frac{A}{2}\left(e^{j\phi}\cdot|H|e^{j\theta}\cdot e^{j\omega_0 n} + e^{-j\phi}\cdot|H|e^{-j\theta}\cdot e^{-j\omega_0 n}\right)$$

$$= \frac{A|H|}{2}\left(e^{j(\phi+\theta+\omega_0 n)} + e^{-j(\phi+\theta+\omega_0 n)}\right) \quad [\because \psi = \omega_0 n + \phi + \theta]$$

$$= A|H|\cdot\frac{1}{2}\left(e^{j\psi} + e^{-j\psi}\right) = A|H|\cos(\psi) = A|H(e^{j\omega_0})|\cos(\omega_0 n + \phi + \theta(\omega_0))$$

$\therefore y_{ss}[n] = A|H(e^{j\omega_0})|\cos[\omega_0 n + \phi + \theta(\omega_0)]$

ω_0 stays the same

Input $A \longrightarrow$ output $A|H(e^{j\omega_0})|$

Input $\phi \longrightarrow$ output $\phi + \theta(\omega_0)$

Response to a causal Exponential Sequence:

Given,

$$x[n] = e^{j\omega n}\mu[n]$$

For a causal LTI system,

$$y[n] = \sum_{K=-\infty}^{\infty} h[K]x[n-K]$$

$$= \sum_{K=-\infty}^{\infty} h[K]\cdot e^{j\omega(n-k)}\cdot\mu[n-K]$$

$$= \sum_{K=0}^{\infty} h[K]\cdot e^{j\omega(n-k)}\cdot\mu[n-k] \quad \left[\begin{matrix}\text{For a causal LTI,}\\ h[K]=0 \text{ for } K<0\end{matrix}\right]$$

Now,

$$\mu[n-K] = \begin{cases} 1 & n-k \geq 0 \\ 0 & n-k<0 \end{cases}$$

$\therefore n-K \geq 0$

or, $K \leq n$

$\therefore$ For $K>n$, $\mu[n-K]=0$

$\therefore$ contributing range $0 \leq K \leq n$

$$= \sum_{K=0}^{n} h[K]\cdot e^{j\omega(n-k)}\cdot\mu[n-k]$$

$$= \sum_{K=0}^{n} h[K]\, e^{j\omega(n-k)}$$

$$= \sum_{K=0}^{n} h[K]\cdot e^{j\omega n}\cdot e^{-j\omega k}$$

$$= e^{j\omega n}\sum_{K=0}^{n} h[K]\cdot e^{-j\omega k}$$

Here, $\sum_{K=0}^{n} h[K]\, e^{-j\omega k} \neq H(e^{j\omega})$ because the upper limit is not ∞

Now, rewriting the sum using an infinite sum,

$$y[n] = e^{j\omega n}\left[\sum_{K=0}^{\infty} h[K]e^{-j\omega k} - \sum_{K=n+1}^{\infty} h[K]\, e^{-j\omega k}\right] \quad \left[\because \sum_{K=0}^{n} = \sum_{K=0}^{\infty} - \sum_{K=n+1}^{\infty}\right]$$

$$= e^{j\omega n}\cdot\sum_{K=0}^{\infty} h[K]\, e^{-j\omega k} - e^{j\omega n}\sum_{K=n+1}^{\infty} h[K]\cdot e^{-j\omega k}$$

$$= H(e^{j\omega})\cdot e^{j\omega n} - e^{j\omega n}\sum_{K=n+1}^{\infty} h[K]e^{-j\omega k}$$

$$y[n] = \underbrace{H(e^{j\omega})\, e^{j\omega n}}_{y_{ss}[n]} \rightarrow \underbrace{e^{j\omega n} \sum_{k=n+1}^{\infty} h[k]\, e^{-j\omega k}}_{y_{tr}[n]}$$

$\therefore y_{ss}[n] = H(e^{j\omega}) \cdot e^{j\omega n}$

$y_{tr}[n] = -e^{j\omega n} \sum_{k=n+1}^{\infty} h[k]\, e^{-j\omega k}$ i.e when $n \to \infty$, sum $\to 0$ and

eventually $y[n] \approx y_{ss}[n]$

The concept of filtering in LTI DTS:

→use $|H(e^{j\omega})|$ to decide which frequencies survive → magnitude response tells us whether a frequency gets passed, reduced, removed, amplified

- $|H(e^{j\omega_0})| \approx 1$ → frequency passes unchanged
- $|H(e^{j\omega_0})| < 1$ → frequency is attenuated
- $|H(e^{j\omega_0})| = 0$ → frequency is removed
- $|H(e^{j\omega_0})| > 1$ → frequency is amplified.

- Passband: range of frequencies filter allows through
- Stopband: range of frequencies filter suppresses

Low pass filter

- passes low frequencies

$|H(e^{j\omega})| \approx 1$ for small $|\omega|$

$|H(e^{j\omega})| \approx 0$ for large $|\omega|$

→ low frequencies pass
high frequencies removed

ex- moving average filter

Highpass filter

- passes only high freq.

$\omega = 0$ for $|H| \approx 0$

ω = larger for larger $|H|$

Bandpass

- mid range of frequencies

→suppresses
very low &
very high frequencies

Band-stop

- low & high frequencies pass, removes mid-range
- called notch filter

tricks

large H near $\omega = 0$ → low pass, & $H \approx 0$ near $\omega = 0$ → high pass

Magnitude response of an ideal loco-pass filter:

$$|H(e^{j\omega})| = \begin{cases} 1 & 0 \le |\omega| \le \omega_c \\ 0 & \omega_c < |\omega| < \pi \end{cases}$$

where, ω_c = cutoff frequency

$|\omega| \le \omega_c \rightarrow$ pass $\omega_c < |\omega| \rightarrow$ stop

— ①

Let,

$$x[n] = A\cos(\omega_1 n) + B\cos(\omega_2 n) \qquad 0 < \omega_1 < \omega_c < \omega_2 < \pi$$

Using linearity,

$$y[n] = A|H(e^{j\omega_1})| \cdot \cos(\omega_1 n + \theta(\omega_1)) + B|H(e^{j\omega_2})| \cdot \cos(\omega_2 n + \theta(\omega_2))$$

Using the thresholds, in ①,

$$y[n] = A|H(e^{j\omega_1})| \cos(\omega_1 n + \theta(\omega_1)) \qquad [\because \omega_1 \le \omega_c \rightarrow |H| = 1,\ \omega_2 > \omega_c \rightarrow |H| = 0]$$

∴ This is a Loco Pass Filter / LPF as

The loco-frequency component survives and higher frequency is removed.

① The input signal consists of a ~~number~~ sum of two cosines sequences of angular frequency 0.1 rad/sample and 0.4 rad/sample. Design a high pass filter (HPF).

For simplicity, assume a FIR filter of length 3; with impulse response,

$$h[0] = h[2] = \alpha_0$$
$$h[1] = \alpha_1$$

$$\therefore y[n] = \sum_{K=\infty}^{\infty} h[K] \cdot x[n-k] = h[0]\cdot x[n] + h[1]\,x[n-1] + h[2]\cdot x[n-2]$$
$$= \alpha_0 x[n] + \alpha_1 x[n-1] + \alpha_0 x[n-2]$$

∴

Our goal is to choose suitable values for filter parameters, α_0 and α_1 so that, the output of the filter is a cosine sequence with frequency of 0.4 rad/sample.

Now, for an FIR:

$$H(e^{j\omega}) = \sum_n h[n] e^{-j\omega n}$$

Since, $n = 0, 1, 2,$

$$\begin{aligned} H(e^{j\omega}) &= h[0] + h[1] e^{-j\omega} + h[2] e^{-j2\omega} \\ &= \alpha_0 + \alpha_1 e^{-j\omega} + \alpha_0 e^{-2j\omega} \\ &= \alpha_0 e^{-j\omega} \cdot e^{j\omega} + \alpha_1 e^{-j\omega} + \alpha_0 e^{-j2\omega} \quad [\because e^{-j\omega} \cdot e^{j\omega} = 1] \\ &= e^{-j\omega}(\alpha_0 e^{j\omega} + \alpha_1 + \alpha_0 e^{-j\omega}) \\ &= e^{-j\omega}[\alpha_0 (e^{j\omega} + e^{-j\omega}) + \alpha_1] \quad [\because e^{-j\omega} + e^{j\omega} = 2\cos\omega] \\ &= e^{-j\omega}(\alpha_1 + 2\alpha_0 \cos\omega) \end{aligned}$$

$\therefore$ Magnitude response, $|H(e^{j\omega})| = |2\alpha_0 \cos\omega + \alpha_1| \cdot \underbrace{|e^{-j\omega}|}_{1}$

$$= |2\alpha_0 \cos\omega + \alpha_1|$$

$e^{j(-\omega)}$

$\therefore$ Phase response, $\theta(\omega) = -\omega + \beta$ where $\beta = 0$ when $2\alpha_0 \cos\omega + \alpha_1 > 0$

$\beta = \pi$ when $2\alpha_0 \cos\omega + \alpha_1 < 0$

Now, we want to design such that, $\omega_1 = 0.1 \rightarrow |H| = 0$

and $\omega_2 = 0.4 \rightarrow |H| = 1$

$\therefore$ $2\alpha_0 \cos(0.1) + \alpha_1 = 0$ — (i)

$2\alpha_0 \cos(0.4) + \alpha_1 = 1$ — (ii)

(ii) − (i),

$2\alpha_0 \cos(0.4) - 2\alpha_0 \cos(0.1) = 1$

or, $\alpha_0 = \dfrac{1}{2[\cos(0.4) - \cos(0.1)]} \approx -6.76$

$\alpha_1 = -2\alpha_0 \cos(0.1)$

$\approx -2(-6.76)\cos(0.1)$

≈ 13.456

$$\therefore y[n] = -6.76\,x[n] + 13.456\,x[n-1] - 6.76\,x[n-2]$$

with $x[n] = \{\cos(0.1n) + \cos(0.4n)\}\,\mu[n]$

Remaining exercises:

4.57

Given,

$$g[n] = \begin{cases} a^n & 0 \le n \le M-1 \\ 0 & \text{otherwise} \end{cases}$$

We know,

$$G(e^{j\omega}) = \sum_{n=0}^{M-1} g[n]\,e^{-j\omega n} = \sum_{n=0}^{M-1} \alpha^n e^{-j\omega n}$$

$$= \sum_{n=0}^{M-1} (\alpha e^{-j\omega})^n$$

$$= \frac{1 - (\alpha e^{-j\omega})^M}{1 - \alpha e^{-j\omega}} \qquad \left[\because \sum_{n=0}^{M-1} p^n = \frac{1-p^M}{1-r};\ r = \alpha e^{-j\omega}\right]$$

$$= \frac{1 - \alpha^M \cdot e^{-j\omega M}}{1 - \alpha e^{-j\omega}}$$

$$= \frac{1}{1 - \alpha e^{-j\omega}} \cdot (1 - \alpha^M e^{-j\omega M}) \qquad \left[\because H(e^{j\omega}) = \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \frac{1}{1-\alpha e^{-j\omega}}\right]$$

$$= H(e^{j\omega})\,(1 - \alpha^M e^{-j\omega M})$$

Now, to find the DC value, putting $\omega = 0$, the current DC value is,

$$G_1(e^{j0}) = \sum_{n=0}^{M-1} \alpha^n = \frac{1-\alpha^M}{1-\alpha}$$

∴ Scale the DC value to 1, divide the whole impulse response by current DC, $c = \frac{1}{G_1(e^{j0})} = \frac{1-\alpha}{1-\alpha^M}$

Scaled response, $g'[n] = cg[n] = \begin{cases} \frac{1-\alpha}{1-\alpha^M} \cdot \alpha^n & 0 \le n \le M-1 \\ 0 & \text{otherwise} \end{cases}$

4.61]

Given, a non-causal FIR:

$$h[n] = a_1\delta[n-3] + a_2\delta[n-2] + a_3\delta[n-1] + a_4\delta[n] + a_5\delta[n+1] + a_6\delta[n+2]$$

Now, we know, $H(e^{j\omega})$ is zero phase or real, for all ω if and only if $h[n]$ is even-symmetric i.e $h[n] = h[-n]$.

Therefore for $h[n]$ to be symmetric;

at $n=3$ and $n=-3$,
$h[3] = h[-3]$
or, $a_1 = 0$

at $n=2$ & $n=-2$,
$h[2] = h[-2]$
$a_2 = a_6$

at $n=1$ & $n=-1$,
$h[1] = h[-1]$
$a_3 = a_5$

at $n=0$, there's no cross-condition, it can be any real valued.

∴ For $a_1=0$, $a_2=a_6$ and $a_3=a_5$ with any arbitrary real value a_4, the given sequence will be zero-phase.

4.63] Given,

$$y[n] = a_1x[n+k] + a_2x[n+k-1] + a_3x[n+k-2] + a_2x[n+k-3] + a_1x[n+k-4]$$

Since,

$$y[n] = \sum h[m]\,x[n-m]$$

and $h[n]$ is defined as the output $y[n]$ when the input is a unit impulse, we can convert as the given difference eqn as

$$h[n] = a_1\delta[n+k] + a_2\delta[n+k-1] + a_3\delta[n+k-2] + a_2\delta[n+k-3] + a_1\delta[n+k-4]$$

Calculating $H(e^{j\omega})$,

$$H(e^{j\omega}) = \sum_n h[n]e^{-j\omega n} = a_1 e^{-j\omega(-k)} + a_2 e^{-j\omega(1-k)} + a_3 e^{-j\omega(2-k)} + a_2 e^{-j\omega(3-k)} + a_1 e^{-j\omega(4-k)}$$

$$= a_1 e^{j\omega k} + a_2 e^{j\omega(k-1)} + a_3 e^{j\omega(k-2)} + a_2 e^{j\omega(k-3)} + a_1 e^{j\omega(k-4)}$$

$$= e^{j\omega k}\left(a_1 + a_2 e^{-j\omega} + a_3 e^{-j2\omega} + a_2 e^{-j3\omega} + a_1 e^{-j4\omega}\right)$$

$$= e^{j\omega k} \cdot e^{-j2\omega}\left(a_1 e^{j2\omega} + a_2 e^{j\omega} + a_3 + a_2 e^{-j\omega} + a_1 e^{-j2\omega}\right)$$

$$= e^{j\omega(k-2)}\left[a_3 + 2a_2\cos\omega + 2a_1\cos 2\omega\right] \quad [\because e^{j\theta} + e^{-j\theta} = 2\cos\theta]$$

For any k, the sum of cosines is a real valued with real coefficients. So, the only term that can make H complex is $e^{-j\omega(k-2)}$ and this factor can be made real i.e equal to 1 when

$$k - 2 = 0$$

or, $k = 2$

4.68 Given,

$h[n] = (0.4)^n \mu[n]$ → causal, stable exponential ($|0.4| < 1$)

Since $h[n] = 0$ for $n < 0$,

Frequency Response, $H(e^{j\omega}) = \sum_{n=0}^{\infty} (0.4)^n e^{-j\omega n} = \sum_{n=0}^{\infty} (0.4e^{-j\omega})^n$

$$= \frac{1}{1 - 0.4e^{-j\omega}} \quad \text{where } \Sigma r^n = \frac{1}{1-r} \text{ if } |r| < 1$$

Now, at $\omega = \frac{\pi}{4}$,

$$H(e^{j\pi/4}) = \frac{1}{1 - 0.4e^{-j\pi/4}} = \frac{1}{1 - 0.4(\cos\frac{\pi}{4} - j\sin\pi/4)} = \frac{1}{1 - 0.4(1/\sqrt{2} - j\,1/\sqrt{2})}$$

$$H(e^{j\frac{\pi}{4}}) = \frac{1}{0.71716 + j0.28284} = 1.20669 - j0.4759$$

$\therefore$ Magnitude, $|H(e^{j\pi/4})| = \sqrt{(1.20669)^2 + (0.4759)^2} = 1.29714$

$\therefore$ Phase, $\theta(\pi/4) = \tan^{-1}\left(\frac{-0.4759}{1.20669}\right) = -21.53^\circ = -0.376$ rad

Now, for any real sequence, $H(e^{-j\omega}) = H^*(e^{j\omega})$. So for $\omega = -\frac{\pi}{4}$

$\therefore H(e^{-j\frac{\pi}{4}}) = 1.20669 + j0.4759$

$\therefore$ Magnitude, $|H(e^{-j\frac{\pi}{4}})| = 1.29714$

$\therefore$ Phase, $\theta(-\frac{\pi}{4}) = \tan^{-1}\left(\frac{0.4759}{1.20669}\right) = 21.53^\circ = 0.376$ rad

Next, given input,

$$x[n] = \sin\left(\frac{\pi n}{4}\right)\mu[n]$$

For an input $x[n] = \sin(\omega_0 n)\mu[n]$, the steady state response/output is given by:

$$y_{ss}[n] = |H(e^{j\omega_0})| \sin(\omega_0 n + \theta(\omega_0))$$

$\therefore$ For $\omega_0 = \frac{\pi}{4}$,

$$y_{ss}[n] = |H(e^{j\pi/4})| \cdot \sin\left(\frac{\pi}{4}n + \theta(\tfrac{\pi}{4})\right)$$

$$= 1.29714 \sin\left(\frac{\pi n}{4} - 21.53^\circ\right)$$

$$= 1.29714 \sin\left(\frac{\pi n}{4} - 0.376\right)$$

4.76] Given,

$$y[n] = d_3 x[n] + d_2 x[n-1] + d_1 x[n-2] + x[n-3] - d_1 y[n-1] - d_2 y[n-2] - d_3 y[n-3]$$

or, $y[n] + d_1 y[n-1] + d_2 y[n-2] + d_3 y[n-3] = d_3 x[n] + d_2 x[n-1] + d_1 x[n-2] + x[n-3]$

Taking fourier transform on both sides,

$$Y(e^{j\omega}) + d_1 e^{-j\omega} Y(e^{j\omega}) + d_2 e^{-j2\omega} Y(e^{j\omega}) + d_3 e^{-j3\omega} Y(e^{j\omega}) = d_3 X(e^{j\omega}) + d_2 e^{-j\omega} X(e^{j\omega}) + d_1 e^{-j2\omega} X(e^{j\omega}) + e^{-j3\omega} X(e^{j\omega})$$

or, $Y(e^{j\omega})[1 + d_1 e^{-j\omega} + d_2 e^{-j2\omega} + d_3 e^{-j3\omega}] = X(e^{j\omega})[d_3 + d_2 e^{-j\omega} + d_1 e^{-j2\omega} + e^{-j3\omega}]$

or, $\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{d_3 + d_2 e^{-j\omega} + d_1 e^{-j2\omega} + e^{-j3\omega}}{1 + d_1 e^{-j\omega} + d_2 e^{-j2\omega} + d_3 e^{-j3\omega}}$

or, $H(e^{j\omega}) = \frac{d_3 + d_2 e^{-j\omega} + d_1 e^{-j2\omega} + e^{-j3\omega}}{1 + d_1 e^{-j\omega} + d_2 e^{-j2\omega} + d_3 e^{-j3\omega}}$ $\left[\because H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}\right]$

which is our frequency response.

Now, let,

$$D(e^{j\omega}) = 1 + d_1 e^{-j\omega} + d_2 e^{-j2\omega} + d_3 e^{-j3\omega}$$

$$\therefore D^*(e^{j\omega}) = 1 + d_1 e^{j\omega} + d_2 e^{j2\omega} + d_3 e^{j3\omega}$$

or, $e^{-j3\omega} \cdot D^*(e^{j\omega}) = e^{-j3\omega} + d_1 e^{-j2\omega} + d_2 e^{-j\omega} + d_3$ → the numerator

Now,

$$H(e^{j\omega}) = \frac{e^{-j3\omega} \cdot D^*(e^{j\omega})}{D(e^{j\omega})} \qquad [\because \left|\frac{A}{B}\right| = \frac{|A|}{|B|} \text{ and } |e^{j\theta}| = 1]$$

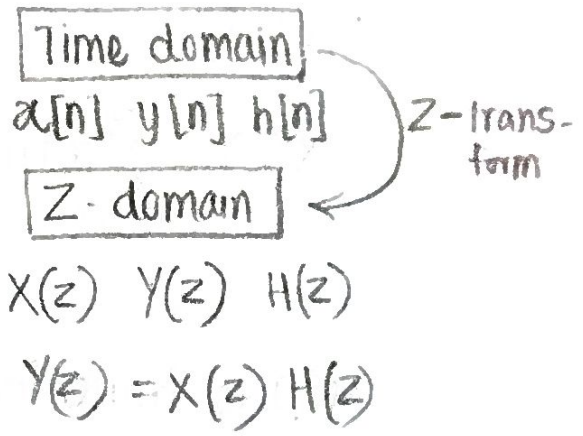
$$\therefore \text{Magnitude, } |H(e^{j\omega})| = \left|\frac{e^{-j3\omega} \cdot D^*(e^{j\omega})}{D(e^{j\omega})}\right| = \frac{|e^{-j3\omega}| \cdot |D^*(e^{j\omega})|}{|D(e^{j\omega})|}$$

$$= \frac{1 \cdot |D^*|}{|D|} = 1 \qquad [\because \text{Magnitude of } D^* = D]$$

for every frequency ω.

CHAPTER - 3
Z - Transform

Why Z-transform?



The DFFT, $X(e^{j\omega}) = \sum_n x[n] e^{-j\omega n}$, its sum doesnt always converge for every sequence (growing exponentials and unstable systems), terms blow up as $n \to \infty$.

z-transform generalizes $e^{j\omega}$ (which is stuck on the unit circle as $|e^{j\omega}| = 1$) to a general complex number z, which can have any magnitude.

- Z transform is more of a general form of DTFT

Z-transform

The z-transform of a discrete time signal $x[n]$ is defined as the power series

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n}$$ where z is a complex variable, $z = re^{j\omega}$

- transforms time-domain signal to complex plane representation $X(z)$
- for convinience,

$X(z) \equiv Z\{x[n]\}$ and $x(n) \xrightarrow{z} X(z)$

→ only exists for those values of z - for which the series converges.

ROC

- set of all values of z for which $X(z)$ is finite

Ex. 3.1.1

@ $x_1[n] = \{\underset{\uparrow}{1}, 2, 5, 7, 0, 1\}$

we know,

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + 0 + z^{-5}$$

$$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$

$X(z)$ is finite for all values except $z = 0$. ∴ ROC = entire z-plane except $z = 0$

(b) $x[n] = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = z^2 + 2z + 5z^0 + 7z^{-1} + 0z^{-2} + z^{-3}$$
$$= z^2 + 2z + 5 + 7z^{-1} + z^{-3}$$

$X(z)$ is finite for all values except $z = \infty$ and $z = 0$.

$\therefore$ ROC = entire z-plane except @ $z = 0$ and $z = \infty$

(c) $x[n] = \delta[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = 1 \cdot z^0 = 1$$

ROC = entire z-plane

(f) $x[n] = \delta[n-k]$ $k > 0$

$\delta[n-k]$ is 1 for $n = k$

$\therefore X(z) = z^{-k}$

ROC = entire z-plane except $z = 0$

(g) $x[n] = \delta[n+k]$ $k > 0$

$X(z) = z^k$

and ROC is the entire z-plane and $z \neq \infty$

Ex 3.1.2 Determine the z-transform of signal, $x[n] = \left(\frac{1}{2}\right)^n u(n)$

We know,

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Here, $x[n]$ is consists of infinite number of non-zero values for $n \geq 0$

$$\therefore X(z) = \sum_{n=0}^{\infty} x[n] z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n \cdot z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2z}\right)^n = \frac{1}{1 - \frac{1}{2} z^{-1}}$$

$$= \frac{z}{z - 0.5}$$

This infinite geometric series converges for $|r| < 1$ or $\left|\left(\frac{1}{2} z^{-1}\right)\right| < 1$

or, $|z| > \frac{1}{2}$ $\therefore$ ROC $\Rightarrow |z| > \frac{1}{2}$

(b) $x[n] = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = z^2 + 2z + 5z^0 + 7z^{-1} + 0z^{-2} + z^{-3}$$
$$= z^2 + 2z + 5 + 7z^{-1} + z^{-3}$$

$X(z)$ is finite for all values except $z = \infty$ and $z = 0$.

$\therefore$ ROC = entire z-plane except @ $z = 0$ and $z = \infty$

(c) $x[n] = \delta[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = 1 \cdot z^0 = 1$$

ROC = entire z-plane

(f) $x[n] = \delta[n-k] \quad k > 0$

$\delta[n-k]$ is 1 for $n = k$

$\therefore X(z) = z^{-k}$

ROC = entire z-plane except $z = 0$

(g) $x[n] = \delta[n+k] \quad k > 0$

$X(z) = z^k$

and ROC is the entire z-plane and $z \neq \infty$

Ex 3.1.2 Determine the z-transform of signal, $x[n] = \left(\frac{1}{2}\right)^n u[n]$

We know,

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Here, $x[n]$ is consists of infinite number of non-zero values for $n \geq 0$

$$\therefore X(z) = \sum_{n=0}^{\infty} x[n] z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n \cdot z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2z}\right)^n = \frac{1}{1 - \frac{1}{2} z^{-1}}$$

$$= \frac{z}{z - 0.5}$$

This infinite geometric series converges for $|r| < 1$ or $\left|\left(\frac{1}{2} z^{-1}\right)\right| < 1$

or, $|z| > \frac{1}{2}$ $\therefore$ ROC $\Rightarrow |z| > \frac{1}{2}$

Ⓟ $x[n] = 2^n u[n]$

$h[n] = 0.5^n u[n]$

Imp for exam

① $y[n] = x[n] * h[n]$

② use frequency domain to calculate

③ use z-transform to calculate

① We know,

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

Now,

$x[k] = 2^k u[k]$ is non-zero for $k \geq 0$

$h[n-k] = 0.5^{n-k} \cdot u[n-k]$ is non-zero for $n \geq k$ or $k \leq n$

So, the sum exists across $0 \leq k \leq n$

$$\therefore y[n] = \sum_{k=0}^{n} 2^k \cdot (0.5)^{n-k} \cdot n \geq 0 = (0.5)^n \sum_{k=0}^{n} 2^k \cdot (0.5)^k = (0.5)^n \sum_{k=0}^{n} 2^k \cdot 2^k$$

$$= (0.5)^n \sum_{k=0}^{n} 4^k$$

$$= (0.5)^n \cdot \frac{1-4^{n+1}}{1-4} = \frac{(0.5)^n (4^{n+1}-1)}{3} = \frac{(0.5)^n \cdot 4^{n+1}}{3} - \frac{(0.5)^n}{3}$$

$$= \frac{4}{3}(0.5 \times 4)^n - \frac{1}{3}(0.5)^n$$

$$= \frac{4}{3}(2)^n - \frac{1}{3}(0.5)^n \quad \text{for } n \geq 0$$

② Frequency domain:

$$y[n] = x[n] * h[n] \longleftrightarrow Y(e^{j\omega}) = X(e^{j\omega})\, H(e^{j\omega})$$

Now,

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} 2^n \cdot u[n] \cdot e^{-j\omega n} = \sum_{n=0}^{\infty} (2e^{-j\omega})^n = \frac{1}{1-2e^{-j\omega}}$$

Similarly, $H(e^{j\omega}) = \sum_{n=0}^{\infty} (0.5)^n e^{-j\omega n} = \sum_{n=0}^{\infty} (0.5e^{-j\omega})^n = \frac{1}{1-0.5e^{-j\omega}}$

$$\therefore Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})$$

$$= \frac{1}{(1-2e^{-j\omega})(1-0.5e^{-j\omega})}$$

③

$$X(z) = \sum_{n=0}^{\infty} 2^n u[n] \cdot z^{-n} = \sum_{n=0}^{\infty} \left(\frac{2}{z}\right)^n = \frac{1}{1-2z^{-1}}, \quad ROC = |z| > 2$$

$$H(z) = \sum_{n=0}^{\infty} (0.5)^n u[n] \cdot z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2z}\right)^n = \frac{1}{1-0.5z^{-1}} \quad ROC = |z| > 0.5$$

$$\therefore Y(z) = \frac{1}{(1-2z^{-1})(1-0.5z^{-1})}$$

where ROC = at least the intersection of the two ROCs, i.e

$|z| > 2$

Now, to invert into original sequence, $y[n]$,

$$Y(z) = \frac{z^2}{(z-2)(z-0.5)}$$

Let,

$$Y(z) \equiv \frac{Az}{z-2} + \frac{Bz}{z-0.5}$$

or, $Y(z)z^2 = Az(z-0.5) + Bz(z-2)$

Putting $z=2$,

$4 = 2A(1.5)$

or, $A = \frac{4}{3}$

Putting $z=0.5$

$0.25 = 0.5B(0.5-2)$

or, $B = -\frac{1}{3}$

$$\therefore Y(z) = \frac{4z}{3(z-2)} - \frac{1z}{3(z-0.5)} = \frac{4}{3} \cdot \frac{1}{1-2z^{-1}} - \frac{1}{3} \frac{1}{1-0.5z^{-1}}$$

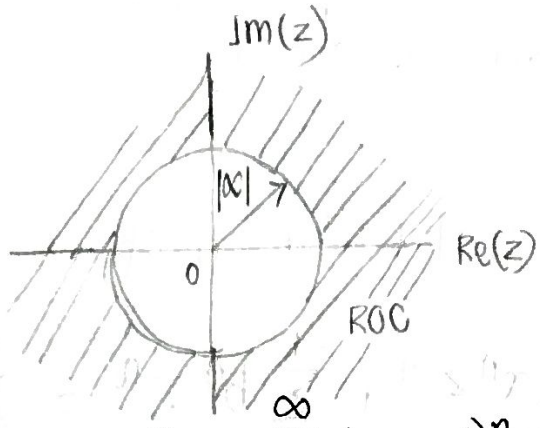
using previously derivide: $\frac{1}{1-az^{-1}} \longleftrightarrow a^n u[n]$,

$$y[n] = \frac{4}{3}(2)^n u[n] - \frac{1}{3}(0.5)^n u[n]$$

Find the z-transform:

Ex 3.1.3

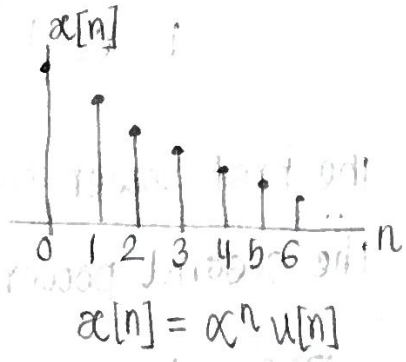
$$x[n] = \alpha^n u[n] = \begin{cases} \alpha^n & n \geq 0 \\ 0 & n < 0 \end{cases}$$



$$X(z) = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n} = \sum_{n=0}^{\infty} \alpha^n u[n] z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n = \frac{1}{1 - z^{-1}\alpha}$$

$$= \frac{z}{z-\alpha}$$

where ROC = $|z| > |\alpha|$ as $|\alpha z^{-1}| < 1$



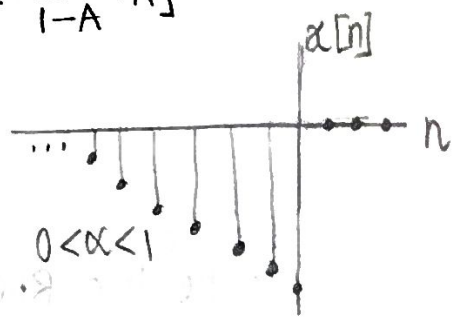
$x[n] = \alpha^n u[n]$

$x[n] = -\alpha^n u[-n-1]$

$$X(z) = \sum_{n=-\infty}^{-1} (-\alpha^n) z^{-n} = -\sum_{l=1}^{\infty} (\alpha^{-1} z)^l \quad [\because l = -n]$$

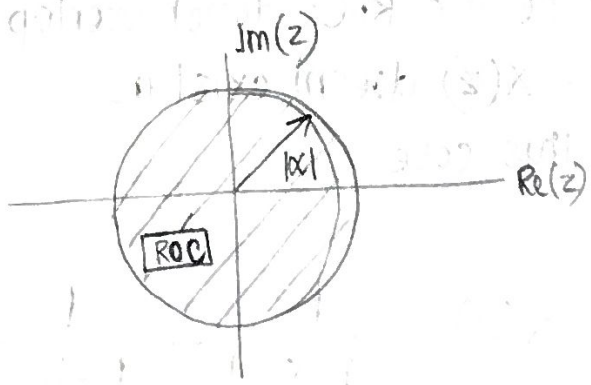
$$= -\frac{\alpha^{-1} z}{1 - \alpha^{-1} z} \quad [\because A + A^2 + A^3 + \cdots = A(1 + A + A^2 + \cdots) = \frac{1}{1-A} \cdot A]$$

$$= \frac{-z\alpha^{-1}}{\frac{\alpha - z}{\alpha}} = \frac{z}{z-\alpha} = \frac{1}{1 - \alpha z^{-1}}$$



Provided that

$|\alpha^{-1} z| < 1$ or $|z| < |\alpha|$



$\therefore$ ROC = $|z| < |\alpha|$

These two examples have the same closed form appearance/expression $\frac{1}{1-\alpha z^{-1}}$. but
This implies that, a closed form expression doesn't identify/uniquely specify the signal in time domain.
The ambiguity is distingushed through ROC.

* A discrete time signal $x[n]$ is uniquely identified by its z-transform $X(z)$ and region of convergence, ROC of $X(z)$

3.1.5

$$x[n] = \alpha^n u[n] + b^n u[-n-1]$$

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} + \sum_{n=-\infty}^{-1} b^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n + \sum_{l=1}^{\infty} (b^{-1}z)^l$$

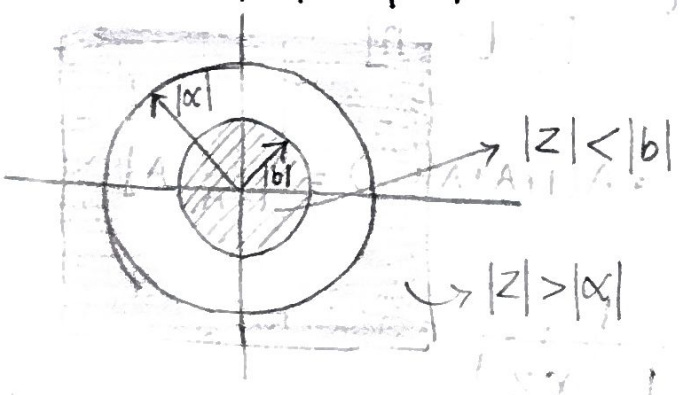
$$= \frac{1}{1-\alpha z^{-1}} - \frac{1}{1-bz^{-1}}$$

$$\frac{b^{-1}z}{1-b^{-1}z} = \frac{z}{b-z} = -\frac{z}{z-b}$$

The first power converges if $|\alpha z^{-1}| < 1$ or $|z| > |\alpha|$
The second power converges if $|b^{-1}z| < 1$ or, $|z| < |b|$

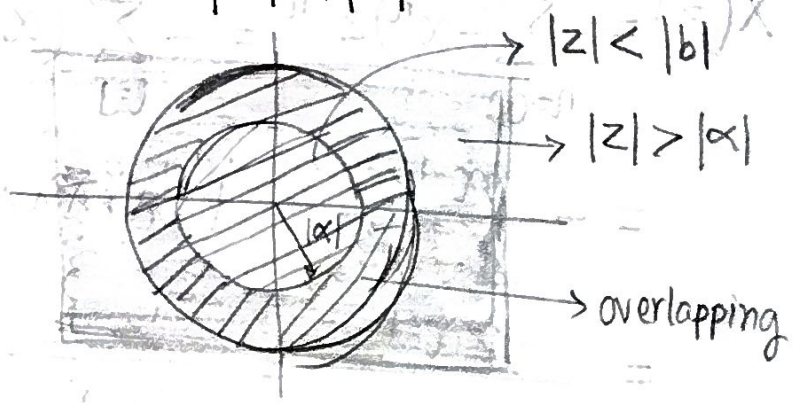
~~ROC = $|z| > |\alpha|$ and $|z| < |b|$~~

Case 1: $|b| < |\alpha|$



- the two ROCs donot overlap
So $X(z)$ doesnt exist in this case

Case 2: $|\alpha| < |b|$



- the two ROCs overlap as a ring as shown in the figure

$\therefore$ ROC = $|\alpha| < |z| < |b|$

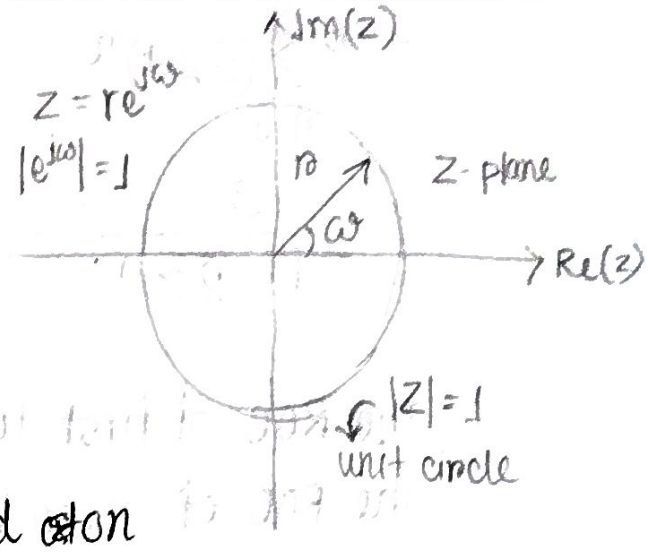
$$\therefore X(z) = \frac{1}{1-\alpha z^{-1}} - \frac{1}{1-bz^{-1}} = \frac{b-\alpha}{\alpha + b - z - \alpha b z^{-1}}$$

with ROC = $|\alpha| < |z| < |b|$ (Ans)

$$x(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x[n] re^{-j\omega n} = x(re^{j\omega}) \quad [\because z = re^{j\omega}]$$

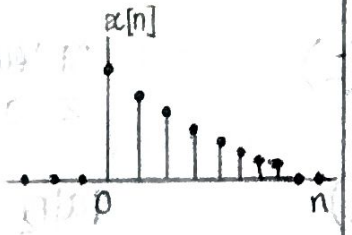
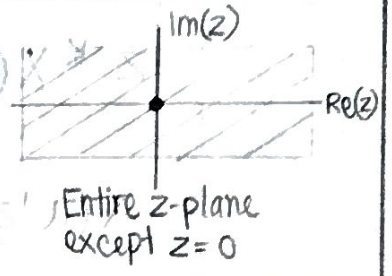
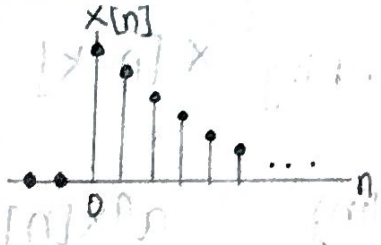
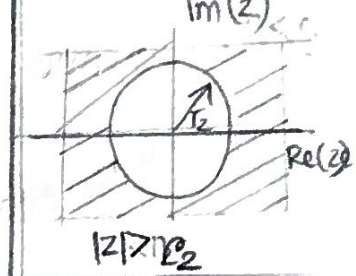
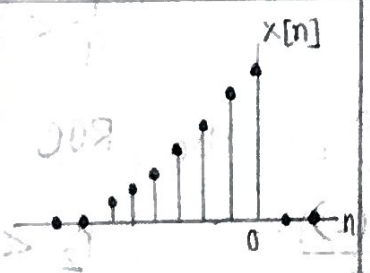
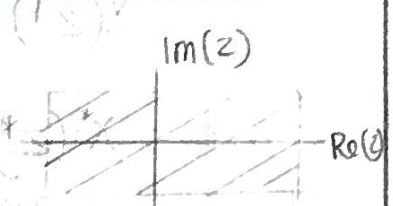
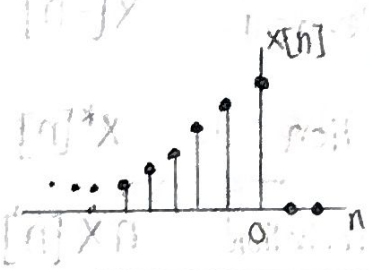
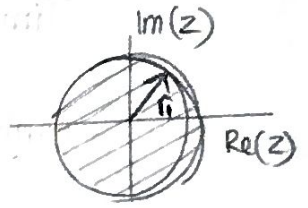
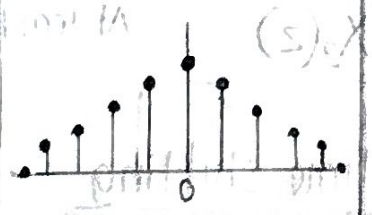
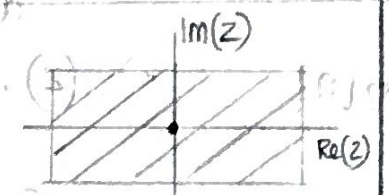
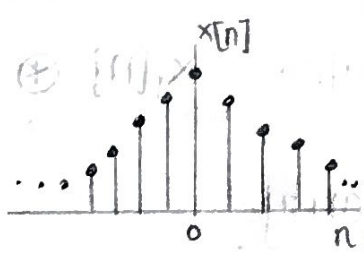
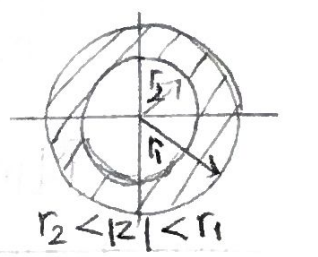
$$\therefore X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \quad [\because r = 1]$$



↳ DTFT formula → DTFT is z-transform evaluated on the unit circle, $r = 1$

* For stable systems ROC will include the unit circle

Characterstic families of signals with their corresponding ROCs:

	Finite Duration Signal		Infinite Duration Signal	
	Signal	ROC	Signal	ROC
Causal		 Entire z-plane except $z = 0$		 $\|z\| > r_2$
Anti-causal		 Entire z-plane except $z = \infty$		 $\|z\| < r_1$
Two-sided		 Entire z-plane except $z = 0$ & $z = \infty$		 $r_2 < \|z\| < r_1$

Ex · 3.2.1

$$x[n] = [3(2)^n - 4(3)^n]\, u[n]$$

$$= 3(2^n) u[n] - 4(3)^n u[n]$$

$$\therefore X(z) = \sum_{n=0}^{\infty} 3(2^n) u[n] z^{-n} - \sum_{n=0}^{\infty} 4(3^n) u[n] z^{-n}$$

$$= 3\sum_{n=0}^{\infty}(2z^{-1})^n - 4\sum_{n=0}^{\infty}(3z^{-1})^n$$

$$= \frac{3}{1-2z^{-1}} - \frac{4}{1-3z^{-1}}$$

The ROC of first term = $|z|>2$

The ROC of second term = $|z|>3$

Intersection of the ROC, ∴ the ROC of $X(z) = |z|>3$

Z-transform properties

Property	Time Domain	Z-Domain	ROC
Linearity	$a_1x_1[n] + a_2x_2[n]$	$a_1X_1(z) + a_2X_2(z)$	At least $ROC_1 \cap ROC_2$
Time-Shifting	$x[n-k]$	$z^{-k}X(z)$	At least ROC except $z=0$ (if $k>0$) & $z=\infty$ (if $k<0$)
Z-Scaling	$a^n x[n]$	$X(a^{-1}z)$	$\lvert a\rvert r_2 < \lvert z\rvert < \lvert a\rvert r_1$
Time-Reversal	$x[-n]$	$X(z^{-1})$	$\frac{1}{r_1} < \lvert z\rvert < \frac{1}{r_2}$
Conjugation	$x^*[n]$	$X^*(z^*)$	ROC
Z-differentiation	$nx[n]$	$-z\frac{dX(z)}{dz}$	$r_2 < \lvert z\rvert < r_1$
Convolution	$x_1[n] \circledast x_2[n]$	$X_1(z)\cdot X_2(z)$	At least $ROC_1 \cap ROC_2$

① Linearity

$x[n] = a_1x_1[n] + a_2x_2[n]$

$\therefore X(z) = \sum_{n=-\infty}^{\infty}(a_1x_1[n] + a_2x_2[n])z^{-n}$

$= \sum_{n=-\infty}^{\infty} a_1x_1[n]\cdot z^{-n} + \sum_{n=-\infty}^{\infty} a_2x_2[n]z^{-n}$

$= a_1\sum_{n=-\infty}^{\infty} x_1[n]z^{-n} + a_2\sum_{n=-\infty}^{\infty} x_2[n]z^{-n}$

$= a_1X_1(z) + a_2X_2(z)$, ROC = $ROC_1 \cap ROC_2$ at least

② Time-shifting

$y[n] = x[n-k]$

$\therefore Y(z) = \sum_{n=-\infty}^{\infty} x[n-k]\cdot z^{-n}$

$= \sum_{m=-\infty}^{\infty} x(m)z^{-(m+k)}$ Let, $m = n-k$ ∴ $n = m+k$

$= \sum_{m=-\infty}^{\infty} x[m]\cdot z^{-m}\cdot z^{-k} = z^{-k}\sum_{m=-\infty}^{\infty} x[m]z^{-m}$

$= z^{-k}\cdot X(z)$

③ Z-Scaling

$$y[n] = a^n x[n]$$

$$\therefore Y(z) = \sum_{n=-\infty}^{\infty} a^n x[n] \cdot z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x[n] \cdot (a^{-1}z)^{-n}$$

$$= X(a^{-1}z)$$

$$ROC = r_1 < |a^{-1}z| \Rightarrow \frac{|z|}{|a|} < r_2$$

$$\therefore |a| r_1 < |z| < |a| r_2$$

Ex. 3.2.2

(a) $x[n] = (\cos\omega_0 n)\, u[n]$

By using Euler's identity,

$$x[n] = \left(\frac{1}{2} e^{j\omega_0 n} + \frac{1}{2} e^{-j\omega_0 n}\right) u[n]$$

$$= \frac{1}{2} e^{j\omega_0 n} \cdot u[n] + \frac{1}{2} e^{-j\omega_0 n} u[n]$$

$$\therefore X(z) = \frac{1}{2} Z\{e^{j\omega_0 n} \cdot u[n]\} + \frac{1}{2} Z\{e^{-j\omega_0 n} \cdot u[n]\}$$

$$= \frac{1}{2}\left(\frac{1}{1 - e^{j\omega_0} z^{-1}}\right) + \frac{1}{2}\left(\frac{1}{1 - e^{-j\omega_0} z^{-1}}\right)$$

where ROC of both terms, $|z| > |e^{\pm j\omega_0}|$

or, $|z| > 1 \quad [\because |e^{j\theta}| = 1]$

$$\therefore X(z) = \frac{1}{2}\left(\frac{1}{1 - e^{+j\omega_0} z^{-1}} + \frac{1}{1 - e^{-j\omega_0} z^{-1}}\right), \quad ROC = |z| > 1$$

(b) $x[n] = (\sin\omega_0 n)\, u[n]$

By using Euler's identity,

$$x[n] = (\sin\omega_0 n) u[n] = \frac{1}{2i}\left[e^{j\omega_0 n} - e^{-j\omega_0 n}\right] u[n]$$

$$\therefore X(z) = \frac{1}{2j} Z\{e^{j\omega_0 n} \cdot u[n]\} - \frac{1}{2j} Z\{e^{-j\omega_0 n} \cdot u[n]\}$$

Using, $a^n u[n] \overset{Z}{\longleftrightarrow} \frac{1}{1 - az^{-1}}$,

$$X(z) = \frac{1}{2j}\left(\frac{1}{1 - e^{j\omega_0} z^{-1}} - \frac{1}{1 - e^{-j\omega_0} z^{-1}}\right)$$

and, $ROC = |z| > |e^{\pm j\omega_0}|$

$$\therefore ROC = |z| > 1$$

$$X(z) = \frac{1}{2j} \frac{1 - e^{-j\omega_0} z^{-1} - 1 + e^{j\omega_0} z^{-1}}{(1 - e^{j\omega_0} z^{-1})(1 - e^{-j\omega_0} z^{-1})}$$

$$= \frac{\frac{1}{2j}(e^{j\omega_0} - e^{-j\omega_0}) z^{-1}}{1 - e^{-j\omega_0} z^{-1} - e^{j\omega_0} z^{-1} + e^{j\omega_0} z^{-1} \cdot e^{-j\omega_0} \cdot z^{-1}}$$

($e^{j\omega_0} z^{-1} \cdot e^{-j\omega_0} \cdot z^{-1} = 1$)

$$= \frac{z^{-1} \cdot \sin\omega_0}{1 - (e^{j\omega_0} + e^{-j\omega_0}) z^{-2}}$$

$$= \frac{z^{-1} \sin\omega_0}{1 - 2\cos\omega_0 \cdot z^{-1} - z^{-2}}$$

Common transform pairs

x[n]	X(z)	ROC	x[n]	X(z)	ROC
$\delta[n]$	1	All z	$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$	$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$	$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert < \lvert a\rvert$
$(\cos\omega_0 n)u[n]$	$\frac{1-z^{-1}\cos\omega_0}{1-2z^{-1}\cos\omega_0+z^{-2}}$	$\lvert z\rvert > 1$	$(\sin\omega_0 n)u[n]$	$\frac{z^{-1}\sin\omega_0}{1-2z^{-1}\cos\omega_0+z^{-2}}$	$\lvert z\rvert > 1$
$a^n(\cos\omega_0 n)u[n]$	$\frac{1-az^{-1}\cos\omega_0}{1-2az^{-1}\cos\omega_0+a^2z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$	$a^n(\sin\omega_0 n)u[n]$	$\frac{az^{-1}\sin\omega_0}{1-2az^{-1}\cos\omega_0+a^2z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$

- Z transforms expressions that are a fraction of polynomials in z^{-1} are called rational. X(z) is a rational function if:

$$X(z) = \frac{B(z)}{A(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{-N}}$$

→ for LTI systems represented by LCCDEs, the z-transform of the unit sample response h[n] denoted by $H(z) = Z\{h(n)\}$ is rational

LCCDE → Linear Constant-Coefficient Difference Eqn

- Zeros: values of z where $X(z) = 0$
- Poles: values of z where $X(z) = \infty$

Poles and zeros of the rational z-transform:

Let $a_0, b_0 \neq 0$

$$\therefore X(z) = \frac{B(z)}{A(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{-N}}$$

$$= \left(\frac{b_0 z^{-M}}{a_0 z^{-M}}\right) \frac{z^M + \left(\frac{b_1}{b_0}\right) z^{M-1} + \cdots + \frac{b_M}{b_0}}{z^N + \frac{a_1}{a_0} z^{N-1} + \cdots + \frac{a_N}{a_0}}$$

$$\therefore X(z) = \frac{b_0}{a_0} z^{-M+N} \frac{(z-z_1)(z-z_2)\cdots(z-z_M)}{(z-p_1)(z-p_2)\cdots(z-p_N)} \quad [\because \text{factorizing}]$$

$$= G_1 z^{N-M} \cdot \frac{\prod_{k=1}^{M}(z-z_k)}{\prod_{k=1}^{N}(z-p_k)}$$

where $z_1, \ldots, z_M$ are finite zeros
$p_1, \ldots, p_K$ are finite poles

→ if $N \neq M$, leftover factor z^{N-M} contributes extra zeros/poles sitting at origin

- $N-M>0$, z^{N-M} +ve power → contributes $|N-M|$ extra zeros at $z=0$ (e.g. $z^3=0$ at $z=0$)
- $N-M<0$, z^{N-M} −ve power → contributes $|N-M|$ extra poles at $z=0$ (e.g. z^{-3})

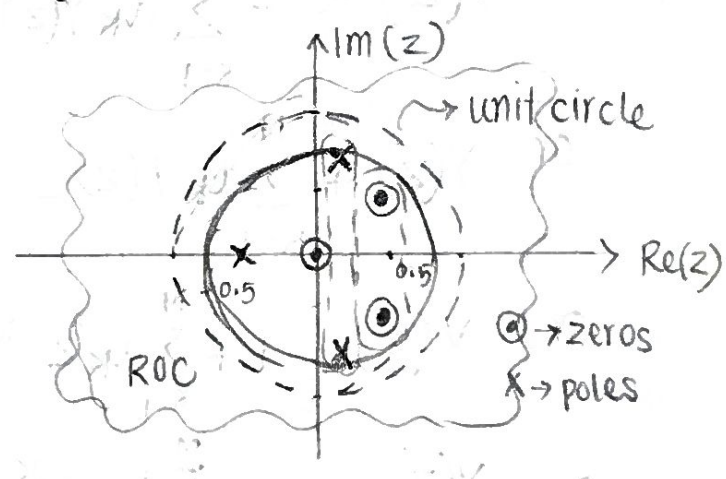
and total no of zeros = total number of poles

Ex- $X(z) = z \cdot \frac{2z^2-2z+1}{16z^3+6z+5}$

$$= \frac{(z-0)\left[z-(0.5+j0.5)\right]\left[z-(0.5-j0.5)\right]}{(z+0.5)\left[z-(0.25+j0.75)\right]\left[z-(0.25-j0.75)\right]}$$

$\therefore$ Poles $= -0.5, \pm 0.25 \pm j0.75$

Zeros $= 0, 0.5 \pm j0.5$



→ complex poles & zeros occur in conjugate but real poles/zeros don't necessarily occur in pairs

→ The ROC is the smallest circle that still encompasses all the poles,
$|z| > |p_{max}| = |z| > |0.25 \pm j0.75|$

Stability from the ROC

- LTI system is stable if the impulse response $h[n]$ is absolutely summable

$$\text{LTI system stable} \longleftrightarrow \sum_{n=-\infty}^{\infty} h[n] < \infty$$

Now,

$$|H(z)| = \left|\sum_{n=-\infty}^{\infty} h[n] z^{-n}\right| \leq \sum_{n=-\infty}^{\infty} \left|h[n] z^{-n}\right| \quad [\because \text{Magnitude of sum is at most the sum of magnitudes}]$$

$$= \sum_{n=-\infty}^{\infty} |h[n]| \cdot |z^{-n}| = \sum_{n=-\infty}^{\infty} |h[n]| \quad \text{for } |z| = 1$$

$\therefore$ LTI system is stable if $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$ and ROC of $H(z)$ contains the unit circle.

→ for stability of a causal ~~sequence~~ system, the poles must lie inside the unit circle.

The System Function

$h[n] \xleftrightarrow{z} H(z)$

impulse response — system function

→ useful representation to find $H(z)$ when the system is described by a LCCDE:

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

$$\therefore Y(z) = -\sum_{k=1}^{N} a_k Y(z) z^{-k} + \sum_{k=0}^{M} b_k X(z) z^{-k}$$

$$\text{or, } Y(z) + \sum_{k=1}^{N} a_k Y(z) z^{-k} = \sum_{k=0}^{M} b_k X(z) z^{-k}$$

$$\text{or, } Y(z)\left[1 + \sum_{k=1}^{N} a_k z^{-k}\right] = \left(\sum_{k=0}^{M} b_k z^{-k}\right) X(z)$$

$$\text{or, } \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = H(z)$$

Rational system function

Ex. 3.3.4 Determine system function & unit sample response:

$$y(n) = \frac{1}{2} y(n-1) + 2x(n)$$

$$Y(z) = \frac{1}{2} Y(z) \cdot z^{-1} + 2X(z) \quad \text{or, } Y(z)\left(1 - \frac{z^{-1}}{2}\right) = 2X(z)$$

$$\text{or, } Y(z) - \frac{1}{2} Y(z) z^{-1} = 2X(z)$$

$$\text{or, } \frac{Y(z)}{X(z)} = \frac{2}{1-\frac{z^{-1}}{2}} = H(z)$$

∴ This system has a pole at $z = \frac{1}{2}$ and zero at the origin

And transforming $H(z) \leftrightarrow h[n]$ through the table $\left[\because a^n u[n] \leftrightarrow \frac{1}{1-az^{-1}}\right]$

$$h[n] = 2 \cdot \left(\frac{1}{2}\right)^n u[n]$$

y[n] = 3x[n] − 4x[n−1] + x[n−2] find h[n].

Taking z-transform on both sides:

$$Y(z) = 3X(z) - 4X(z) \cdot z^{-1} + X(z) \cdot z^{-2}$$

$$\text{or, } Y(z) = X(z)\left[3 - 4z^{-1} + z^{-2}\right]$$

$$\text{or, } \frac{Y(z)}{X(z)} = 3 - 4z^{-1} + z^{-2}$$

$$\text{or, } H(z) = 3 - 4z^{-1} + z^{-2}$$

Transforming $H(z) \leftrightarrow h[n]$ using $\delta[n] \overset{z}{\leftrightarrow} 1$ and $x[n-k] \overset{z}{\leftrightarrow} z^{-k}X(z)$:

$$h[n] = 3\delta[n] - 4\delta[n-1] + \delta[n-2]$$

Inversion of z-transform

Three methods:

(i) Direct Power series Expansion (ii) Contour Integration (iii) Partial fraction & table look-up

$$X(z) = \sum_{n=-\infty}^{\infty} x[k] z^{-k} \qquad x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

(i) Expansion into power series:

→ z-transform is defined as a power series i.e $X(z) = \sum_k x[k] z^{-k}$ whose co-efficients are the sequence values. So, given $X(z)$ is expanded into exact power-series shape → read $x[k]$ with whatever co-efficient is multiplied with z^{-k}

$X(z) = \log(1+az^{-1}) \quad |z| > |a|$

From the Taylor series for $\log(1+u)$:

$$\log(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} u^n$$

Substituting $u = az^{-1}$,

$$X(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cdot (az^{-1})^n = \sum_{n=1}^{\infty} \left[\frac{(-1)^{n+1} a^n}{n}\right] z^{-n}$$

By matching co-efficients of z^{-n} term by term:

$$x[n] = \begin{cases} \frac{(-1)^{n+1} a^n}{n} & n \geq 1 \\ 0 & n \leq 0 \end{cases}$$

#use this method when series expansion can be used (log, exponents, sin/cos, geometric)

② Partial Fraction

- most used

Steps:

1. Find distinct poles of $X(z)$: $P_1, P_2 \ldots, P_K$ and corresponding multiplicities $m_1, m_2, \ldots m_K$

2. Partial fraction is of form:

$$z^{-R} X(z) = \sum_{K=1}^{K} \left(\frac{A_{1K}}{z - P_K} + \frac{A_{2K}}{(z-P_K)^2} + \cdots + \frac{A_{mK}}{(z-P_K)^{m_K}} \right)$$

where, P_K is the m_K^{th} order pole and R is selected to make a strictly proper rational function

3. Use appropiate approach to calculate $\{A_{iK}\}$

Ex. Find $x[n]$, given poles of $X(z)$ at $P_1 = -2$ and double pole at $P_2 = P_3 = 1$.

$$X(z) = \frac{1}{(1+2z^{-1})(1-z^{-1})^2} \quad \rightarrow \text{given}$$

$$\frac{X(z)}{z} = \frac{z^2}{(z+2)(z-1)^2}$$

Let,

$$\frac{X(z)}{z} \equiv \frac{A}{z+2} + \frac{B}{(z-1)^2} + \frac{C}{(z-1)^2}$$

or, $$\frac{z^2}{(z+2)(z-1)^2} \equiv \frac{A}{z+2} + \frac{B}{z-1} + \frac{C}{(z-1)^2}$$

or, $$z^2 = A(z-1)^2 + B(z-1)(z+2) + C(z+2)$$

Putting $z=1$, $z=-2$, Equating co-efficients of z^2,

or, $1 = 0+0+3C$ $\quad 4 = 9A+0+0$ $\quad 1 = A+B$

$\therefore C = \frac{1}{3}$ $\quad A = \frac{4}{9}$ $\quad$ or, $1 = \frac{4}{9} + B \quad \therefore B = \frac{5}{9}$

$$\therefore \frac{X(z)}{z} = \frac{4}{9(z+2)} + \frac{5}{9(z-1)} + \frac{1}{3(z-1)^2}$$

$$\therefore X(z) = \frac{4z}{9(z+2)} + \frac{5z}{9(z-1)} + \frac{z}{3(z-1)^2}$$

$$= \frac{4}{9(1+2z^{-1})} + \frac{5}{9(1-z^{-1})} + \frac{z^{-1}}{3(1-z^{-1})^2}$$ [Dividing each term by power of z matching poles]

$$= \frac{4}{9} \cdot \frac{1}{(1+2z^{-1})} + \frac{5}{9} \cdot \frac{1}{(1-z^{-1})} + \frac{1}{3} \frac{z^{-1}}{(1-z^{-1})^2}$$

Converting each term using $a^n u[n] \leftrightarrow \frac{1}{1-az^{-1}}$; (assuming causality)

$$x[n] = \frac{4}{9}(2)^n u[n] + \frac{5}{9} u[n] + \frac{n}{3} u[n]$$

CHAPTER 4
Frequency Analysis of Signals

- Periodic components : complicated signals be built/ decomposed into sinusoids
- Harmonics : signal repeats with period T_0, $\omega_0 = \frac{2\pi}{T_0}$ the only frequencies allowed to appear in its decomposition are multiples of ω_0
  $\omega_0, 2\omega_0, 3\omega_0, \ldots \rightarrow$ 1st, 2nd, 3rd ... harmonics
- Resonance : systems tendency to respond with much larger amplitude when driven at specific frequencies
- Bandwidth : range of frequencies over which a signals energy/power is concentrated.

} Frequency Analysis : resolution of signal into its frequency components

Tools for frequency analysis :

	Period	Aperiodic
Continuous Time Signal	Fourier Series	Fourier Transform (CTFT)
Discrete Time Signal	Discrete Time Fourier Series	Discrete Time Fourier Transform (DTFT) ~ z-transform

- sinusoidal carrier → periodic
- ECG → aperiodic
- Speech → aperiodic
- Vibration → periodic
- Finite Sensor Record → periodic

CT-fourier Series

$$x(t) = \sum_{K=-\infty}^{\infty} c_K e^{jk\omega_0 t}$$

c_K = Fourier Series Coefficient

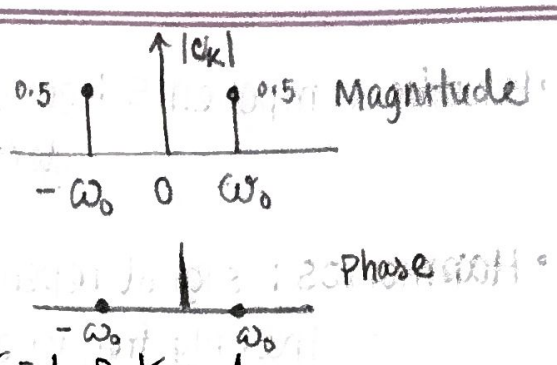
- any periodic signal $x(t)$ can be with period T_0 and fundamental frequency ω_0 can be reconstructed as a weighted sum of complex exponentials at harmonic frequency $k\omega_0$ for every integer k.

The weights c_k tells how much of each harmonic is present.

Ex-

① $x(t) = \cos\omega_0 t$

$= \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}$



Matching term by term with $\sum_{K=-\infty}^{\infty} c_k e^{jk\omega_0 t}$, $K=1$ & $K=-1$,

$c_1 = \frac{1}{2}$, $c_{-1} = \frac{1}{2}$, $c_K = 0$ for all other k

At $K=0$, $c_0 = 0$ ∴ The DC component (constant, non-varying part) = 0.

② $x(t) = A\cos(\omega_0 t + \phi)$

$= \frac{1}{2} A e^{j(\omega_0 t+\phi)} + \frac{1}{2} A e^{-j(\omega_0 t+\phi)}$ $\quad [\because \cos\omega_0 t = \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}]$

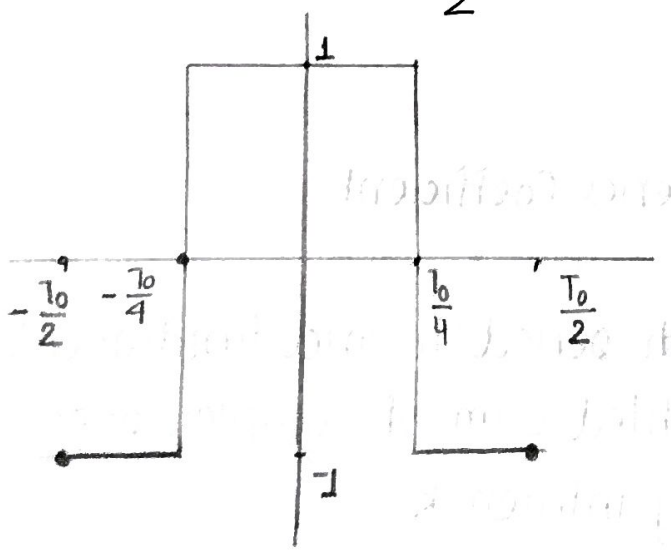
$= \frac{1}{2}\left(Ae^{j\phi}\right) e^{j\omega_0 t} + \frac{1}{2}\left(Ae^{-j\phi}\right) e^{-j\omega_0 t}$

$\therefore c_1 = \frac{1}{2} Ae^{j\phi}$

$c_{-1} = \frac{1}{2} Ae^{-j\phi}$

} $c_{-1} = c_1^*$ ⇝ conjugate symmetry

③ $x(t) = \begin{cases} 1 & -\frac{T_0}{4} < t < \frac{T_0}{4} \\ -1 & -\frac{T_0}{2} < t < -\frac{T_0}{4} \text{ and } \frac{T_0}{4} < t < \frac{T_0}{2} \end{cases}$



$c_K = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) \cos(K\omega_0 t)\, dt$ ←

$c_K = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{T_0/2} x(t)\, e^{-jk\omega_0 t}\, dt$

$= \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) \cdot (\cos k\omega_0 t - j\sin k\omega_0 t)\, dt$

Since, $x(t)$ is even, i.e symmetric about $t = 0$ and $\sin k\omega_0 t$ is odd, product $x(t) \cdot \sin(k\omega_0 t)$ is odd. Integral of any odd function over a period symmetric interval is zero.

$$\Rightarrow c_K = \frac{2}{T_0}\int_0^{\frac{T_0}{2}} x(t)\cos(k\omega_0 t)\,dt \quad [\because \text{The left half mirrors the right, as the function is even}]$$

$$= \frac{2}{T_0}\int_0^{\frac{T_0}{4}} x(t)\cos(k\omega_0 t)\,dt + \frac{2}{T_0}\int_{\frac{T_0}{4}}^{\frac{T_0}{2}} x(t)\cos(k\omega_0 t)\,dt \quad [\because x(t) \text{ takes different values at two intervals}]$$

$$= \frac{2}{T_0}\left[\int_0^{\frac{T_0}{4}} (1)\cos(k\omega_0 t)\,dt + \int_{\frac{T_0}{4}}^{\frac{T_0}{2}} (-1)\cos(k\omega_0 t)\,dt\right]$$

$$= \frac{2}{T_0}\left[\frac{\sin(k\omega_0 t)}{k\omega_0}\Big|_0^{\frac{T_0}{4}} - \frac{\sin(k\omega_0 t)}{k\omega_0}\Big|_{\frac{T_0}{4}}^{\frac{T_0}{2}}\right]$$

$$= \frac{2}{T_0}\left[\left(\frac{\sin\left(\frac{2\pi k}{T_0}\cdot\frac{T_0}{4}\right)}{k\omega_0} - 0\right) - \left(\frac{\sin\left(\frac{2\pi k}{T_0}\cdot\frac{T_0}{2}\right)}{k\omega_0} - \frac{\sin\left(\frac{2\pi k}{T_0}\cdot\frac{T_0}{4}\right)}{k\omega_0}\right)\right] \qquad \sin k\pi = 0$$

$$= \frac{2}{T_0}\left[\frac{\sin\frac{k\pi}{2}}{k\omega_0} - 0 + \frac{\sin k\pi/2}{k\omega_0}\right]$$

$$= \frac{4\sin(k\pi/2)}{k\omega_0 \cdot T_0} = \frac{4\sin(k\pi/2)}{\frac{2\pi k}{T_0}\cdot T_0} = \frac{2}{k\pi}\sin\left(\frac{k\pi}{2}\right)$$

Now, at $k=0$, $c_0 = 0$

and, for even k, $(\pm 2, \pm 4, \pm 6, \ldots)$: $\sin\frac{k\pi}{2} = 0$, $c_k = 0$

for odd k, $(\pm 1, \pm 3, \pm 5, \ldots)$: $\sin\frac{k\pi}{2} = \pm 1$, $c_k = \pm\frac{2}{k\pi}$

$\therefore$ magnitude, $= \frac{2}{\pi|k|}$

Only odd harmonics survive, decreasing as $\frac{1}{K}$ for $K \to \infty$

* For a square wave with +1 and 0 $\rightarrow$ $c_0 = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t)\,dt = \frac{1}{T_0}\int_{\frac{-T_0}{2}}^{\frac{T_0}{2}} (1)\,dt = \cdot^{1}$

and $c_K = \frac{1}{k\pi}\sin\left(\frac{\pi k}{2}\right)$

$= \frac{1}{T_0}\cdot 1\cdot\left(\frac{T_0}{4} - \left(-\frac{T_0}{4}\right)\right)^{\frac{T_0}{2}} = \frac{1}{T_0}\cdot\frac{T_0}{2} = \frac{1}{2}$

CTFT

For a continuous time aperiodic signal, $x(t)$, the Continuous Time Fourier Transform pair is: $\Omega \rightarrow$ radial frequency : rad/s

- Analysis Equation (forward transform):
  - finds the continuous spectrum, $X(\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t}\, dt$
- Synthesis Equation (inverse transform):
  - reconstructs $x(t)$ from the spectrum, $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega)\, e^{j\Omega t}\, d\Omega$
  - $\rightarrow$ synthesizes time domain signal from its frequency content.

▶ Using cyclic frequency, $2\pi F$:

- Synthesis : $x(t) = \int_{-\infty}^{\infty} X(F)\, e^{j2\pi Ft}\, dt$
- Analysis : $X(F) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi Ft}\, dF$

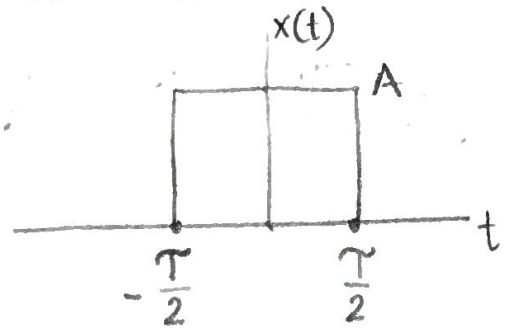
*Dirichlet Conditions:

A continuous signal $x(t)$ has CTFT, by allowing $T_p \rightarrow \infty$ in CTFS:

1. $x(t)$ has finite number of finite discontinuities
2. $x(t)$ has a finite number of maxima & minima
3. $x(t)$ is absolutely integrable: $\int_{-\infty}^{\infty} |x(t)|\, dt < \infty$

Ex. 4.1.2

Determine fourier transform of the following: $x(t) = \begin{cases} A & |t| < T/2 \\ 0 & |t| > T/2 \end{cases}$



signal is aperiodic and satifies dirichlet:

$$X(\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t} = \int_{-T/2}^{T/2} A\, e^{-j\Omega t}\, dt$$

$$= \frac{-Ae^{-j\Omega t}}{j\Omega}\Big|_{-T/2}^{\frac{T}{2}} = -\frac{A}{j\Omega}\left(e^{-j\Omega T/2} - e^{+j\Omega T/2}\right)$$

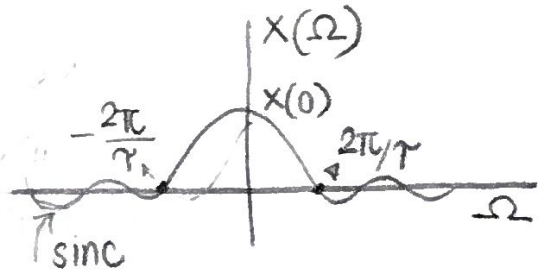
$$= -\frac{A}{j\Omega}\left(-2j\sin\left(\frac{\Omega\tau}{2}\right)\right)$$

$$= \frac{2A\sin\left(\frac{\Omega\tau}{2}\right)}{\Omega}$$

Since, $F = \frac{\Omega}{2\pi}$, $X(F) = \frac{2A\sin\left((2\pi F)\tau/2\right)}{2\pi F} = \frac{A\sin(\pi F\tau)}{\pi F}$

we know, $\text{sinc}(x) = \frac{\sin \pi x}{\pi x}$

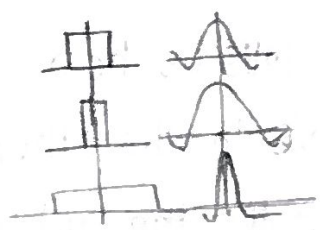
$\therefore X(F) = \frac{A\tau \sin(\pi F\tau)}{\pi F\tau} = A\tau\,\text{sinc}(F\tau)$



Here, τ = pulse width,

$\therefore$ At $F=0$, $X(0) = A\tau$ → area of rectangular pulse width

$X(F) = 0$ when $F = \frac{1}{\tau}$ → wider the pulse, the narrower the sinc and vice versa.

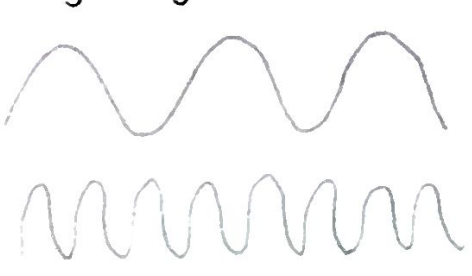


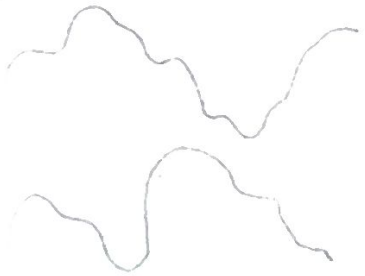
CTFT Intuition

$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega)e^{j\Omega t}d\Omega$ says $x(t)$ is a linear combination of $e^{j\Omega t}$ for every $\Omega \in \mathbb{R}$

i.e. the larger the $|X(\Omega)|$, the more $x(t)$ will look like a pure sinusoid at frequency Ω.

→ $X(\Omega)$ is literally the sum of weight that carries frequency Ω in the reconstruction sum. If one particular Ω has much larger $|X(\Omega)|$, that single frequency dominates the sum, $x(t)$ looks like an almost clean sinusoid at that dominant frequency with everything else as minor texture/noise.





$$= -\frac{A}{j\Omega}\left(-2j\sin\left(\frac{\Omega\tau}{2}\right)\right)$$

$$= \frac{2A\sin\left(\frac{\Omega\tau}{2}\right)}{\Omega}$$

Since, $F = \frac{\Omega}{2\pi}$, $X(F) = \frac{2A\sin\left((2\pi F)\tau/2\right)}{2\pi F} = \frac{A\sin(\pi F\tau)}{\pi F}$

we know, $\text{sinc}(x) = \frac{\sin \pi x}{\pi x}$

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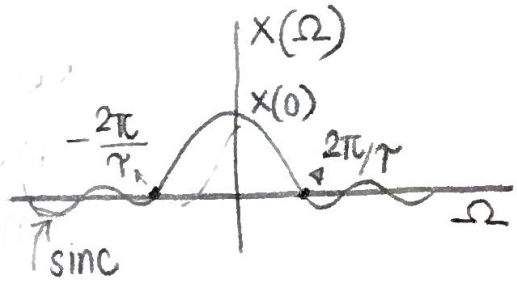

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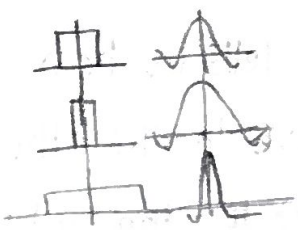

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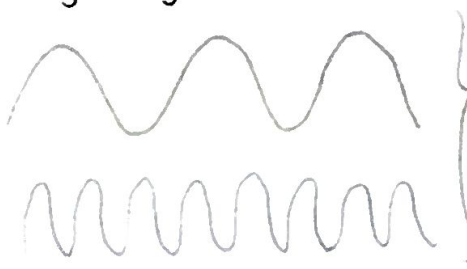
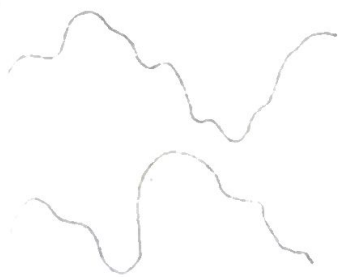




every Ω competes to look like a sinusoid, no dominant peak, $|X(\Omega)|$ roughly spread out

CTFT Duality:

$$\blacktriangleright\ x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega)e^{j\Omega t}\,d\Omega \qquad \blacktriangleright\ X(\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}\,dt$$

- two formulas are almost same, with the roles of t & Ω swapped

Thus near-symmetry also produces:

Time Domain		Frequency Domain
Rectangle	$\overset{\mathcal{F}}{\longleftrightarrow}$	sinc
Sinc	$\overset{\mathcal{F}}{\longleftrightarrow}$	rectangle
convolution	$\overset{\mathcal{F}}{\longleftrightarrow}$	multiplication
multiplication	$\overset{\mathcal{F}}{\longleftrightarrow}$	convolution

CTFT: Magnitude and Phase

→ Phase contains most information

- Magnitude tells us how much of each frequency is present, but phase tells us 'when' relative to the others, each of those frequency components peaks.

→ Phase encodes the actual timing/alignment of all sinusoidal components relative to each other.

→ same magnitude + scrambled phase → all recognizable structures, (audio, edges, shapes) destroyed

* So when filtering, preserve phase no matter the condition.

↳ ex) linear phase filters: all frequencies shifted in time by same amount keeping the relative phase relationship intact.

Frequency Analysis of Discrete Time Signals

Fourier series for DT Periodic Signal

- Synthesis: $x[n] = \sum_{K=0}^{N-1} c_K e^{j2\pi kn/N}$ → often called DTFS
- Analysis: $c_K = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$

where, sequence $x[n] = x[n+N]$ & N is the period

$c_K = c_{K+N}$ is a periodic sequence with fundamental period N

For a sampling frequency F_s, range $0 \le K \le N-1$ corresponds to $0 \le F \le F_s$

$\frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-j2\pi(K+N)n/N} = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} e^{-j2\pi n}$ (always 1 for int n) $= c_K$

$\therefore c_{K+N} = c_K$

Ex - 4.2.1

Determine spectra of signals:

ⓐ $x[n] = \cos\sqrt{2}\pi n$

$\omega_0 = \sqrt{2}\pi$

$\therefore f_0 = \frac{1}{\sqrt{2}}$

which is not a rational number, signal is aperiodic.

Spectral content consists of single frequency component $\omega = \omega_0 = \sqrt{2}\pi$

ⓑ $x[n] = \cos\left(\frac{\pi n}{3}\right)$

$\omega_0 = \frac{\pi}{3}$, $f_0 = \frac{1}{6}$ $\therefore N = 6$

$c_K = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$

$= \frac{1}{6}\sum_{n=0}^{5} \cos\left(\frac{\pi n}{3}\right) \cdot e^{-j2\pi kn/6}$ $K = 0, 1, \ldots, 5$

$= \frac{1}{6}\sum_{n=0}^{5} \left(\frac{1}{2}e^{j2\pi n/6} + \frac{1}{2}e^{-j2\pi n/6}\right) e^{-j2\pi kn/6}$

But $\cos\left(\frac{\pi n}{3}\right) = \cos\left(\frac{2\pi n}{6}\right) = \frac{1}{2}e^{j2\pi n/6} + \frac{1}{2}e^{-j2\pi n/6}$

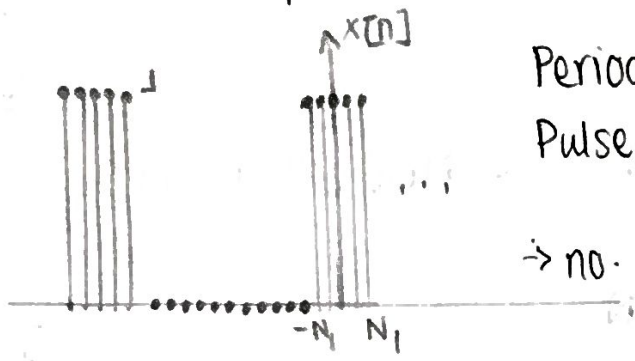
which is in form of c_K.

$\therefore c_1 = \frac{1}{2}$, $c_5 = \frac{1}{2}$

$c_0 = c_2 = c_3 = c_4 = 0$

$e^{-j2\pi n/6} = e^{j2\pi(5-6)n/6} = e^{j2\pi(5n)/6}$ → $K = 5$

*DTFS on a square wave with period N:



Period = N

Pulse width = each signal is equal to 1 for $n = -N_1$ to N_1 and 0 elsewhere

→ no. of non-zero samples in one period $2N_1+1$

Acc to analysis eqn:

$$c_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-j(2\pi/N)kn}$$

Because $x[n] = 1$ only for $-N_1$ to N_1, $c_K = \frac{1}{N} \sum_{n=-N_1}^{N_1} e^{-jk(2\pi/N)n}$

- Case 1: $k = 0, \pm N, \pm 2N \ldots$ (DC and harmonics)

when $k=0$, exponential, $e^0 = 1$, sum counts the number of samples.

$$c_0 = \frac{1}{N} \sum_{n=-N_1}^{N_1} 1 = \frac{2N_1+1}{N}$$

$K = mN$

$e^{-jk(2\pi/N)n} = e^{-j(mN)2\pi/N \cdot n}$

$= e^{-jm2\pi n} = e^0 = 1$

This is the average value of square wave i.e. the DC component.

- Case 2: $k \neq 0, \pm N, \pm 2N \ldots$

$$c_K = \frac{1}{N} \cdot \frac{e^{jk(2\pi/N)N_1} - e^{-jk(2\pi/N)(N_1+1)}}{1 - e^{-jk(2\pi/N)}}$$

$$= \frac{1}{N} \cdot \frac{e^{j\alpha N_1} e^{j\alpha/2} - e^{-j\alpha(N_1+1)} e^{j\alpha/2}}{\left(1 - e^{-j\alpha}\right) e^{j\alpha/2}}$$

[∵ Let, $\alpha = \frac{2\pi k}{N}$ & multiplying num-denom by $e^{-j\alpha/2}$]

$$= \frac{1}{N} \cdot \frac{e^{j\alpha(N_1+\frac{1}{2})} - e^{-j\alpha(N_1+1/2)}}{e^{j\alpha/2} - e^{-j\alpha/2}}$$

$$= \frac{1}{N} \cdot \frac{2j \sin\left(\alpha(N_1+\frac{1}{2})\right)}{2j \sin\left(\frac{\alpha}{2}\right)} = \frac{\sin\left(\alpha(N_1+\frac{1}{2})\right)}{\sin\left(\frac{\alpha}{2}\right)} = \frac{\sin\left(\frac{2\pi k(N_1+\frac{1}{2})}{N}\right)}{\sin\left(\frac{\pi k}{N}\right)}$$

Power density spectrum of periodic signal

The average power of a periodic signal in time domain,

$$P_x = \frac{1}{N}\sum_{n=0}^{N-1} |x[n]|^2$$

Substituting the synthesis eqn, $x[n] = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$

$$P_x = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\cdot x^*[n]$$

$$= \frac{1}{N}\sum_{n=0}^{N-1} x[n]\cdot\left(\sum_{k=0}^{N-1} c_k^* e^{-j2\pi kn/N}\right)$$

$$= \sum_{k=0}^{N-1} c_k^* \underbrace{\left[\frac{1}{N}\sum_{n=0}^{N-1} x[n]\cdot e^{-j2\pi kn/N}\right]}_{\text{Analysis eqn, } c_k}$$

$$\therefore P_k = \sum_{k=0}^{N-1} c_k^* \cdot c_k$$

$$= \sum_{k=0}^{N-1} |c_k|^2$$

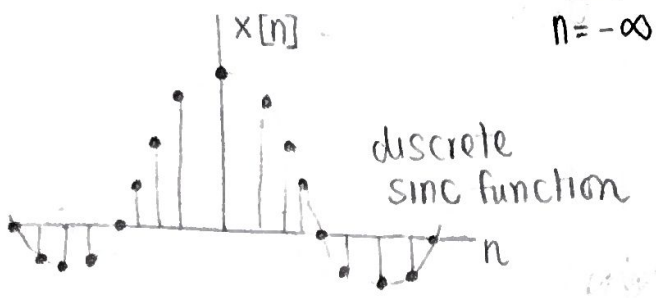
i.e the total power of a signal is the sum of the powers of each harmonic. This shows how the total power is distributed across different frequencies.

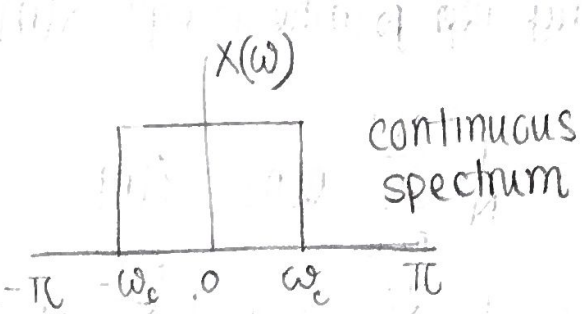
* If $x[n]$ is a real signal, $x^*[n] = x[n]$. So, $c_k^* = c_k$.

So, due to this symmetry, if coefficients for $k = 1, 2, \ldots N/2$ are known, the rest half is also known, no need to calculate the other half of spectrum.

DTFT for aperiodic signals

- Synthesis eqn: $x[n] = \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{j\omega n} d\omega$
- Analysis eqn: $X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$





- unlike CTFT, the DTFT spectrum $X(\omega)$ is always periodic with period 2π

i.e $X(\omega+2\pi) = X(\omega)$ because $e^{-j(\omega+2\pi)n} = e^{-j\omega n} \cdot e^{-j2\pi n} = e^{-j\omega n}$

Energy Density Spectrum

Energy in Time domain,

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \sum_{n=-\infty}^{\infty} x[n] \cdot x^*[n]$$

Substituting the synthesis eqn,

$$E_x = \sum_{n=-\infty}^{\infty} x[n] \cdot \left[\frac{1}{2\pi} \int_{2\pi} X^*(\omega) e^{-j\omega n}\right]$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(\omega) \left[\sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}\right] d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(\omega) \cdot X(\omega)$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega \qquad S_{xx}(\omega) = |X(\omega)|^2$$

The term, $|X(\omega)|^2$ is the Energy Density Spectrum, denoted as $S_{xx}(\omega)$
Tells us how much energy is packed into each tiny sliver of frequency

→ For real signals, spectrum has perfect mirror symmetry

∴ Even magnitude, $|X(\omega)| = |X(-\omega)|$ ⇝ magnitude of neg frequency = +ve Freq.

Even energy density, $S_{xx}(-\omega) = S_{xx}(\omega)$ → energy density perfectly symmetric around 0.

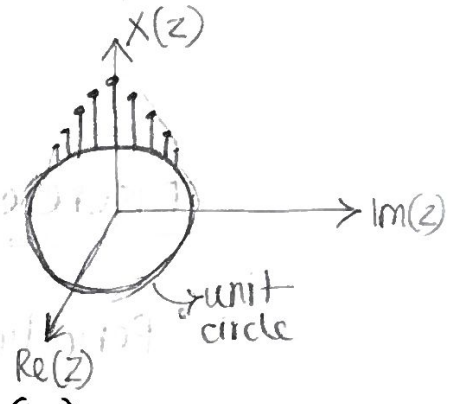
Odd phase, $\angle X(-\omega) = -\angle X(\omega)$ → phase is antisymmetric

∴ Frequency range for real signals $0 \leq \omega \leq \pi$ or $0 \leq F \leq F_s/2$

Relationship of fourier transform to z-transform

z-transform, $X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$ ROC : $r_2 \leq |z| \leq r_1$

Putting $z = re^{j\omega}$, $X(z)|_{z=re^{j\omega}} = \sum [x[n] \cdot r^n] e^{-j\omega n}$



If $X(z)$ converges for $|z| = 1$, $X(z) = \sum x[n] e^{-j\omega n} = X(\omega)$

∴ Fourier transform is viewed as the z-transform of the sequence evaluated on the unit circle.

Frequency Domain Classifications : Bandwidth

Three broad catagories based on frequency:

1. Low-frequency signal: energy bunched up around 0 ex- slow wave / DC voltage
2. High frequency signal: energy around the high end, near π or $\frac{F_s}{2}$
3. Medium frequency / Bandpass signal: energy concentrated at somewhere in the middle, ex - radio station

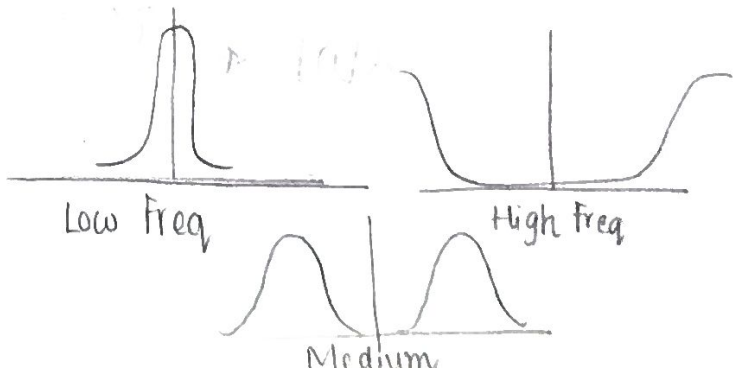
- Bandwidth : Difference of highest & lowest frequency, quantitative measure of range of frequencies over which power/energy concentrated

Three catagories:

① Narrowband : - very thin spike

② Wideband : broad, sprawling hill

③ Bandlimited : signal is zero above a certain frequency



Frequency domain & Time domain properties

Continuous time

- Periodic → F.D: Discrete, Aperiodic — Fourier Series

$$c_K = \frac{1}{T_P}\int_{T_P} x(t)e^{-2\pi k F_0 t}\,dt$$

$$x(t) = \sum_{K=-\infty}^{\infty} c_K e^{j2\pi k F_0 t}$$

- Aperiodic → Continuous Aperiodic — Fourier Transform

$$X(F) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi Ft}\,dt$$

$$x(t) = \int_{-\infty}^{\infty} X_a(F) e^{-j2\pi Ft}\,dF$$

Discrete Time

- Periodic → F.D: Discrete, Periodic — Fourier Series

$$c_K = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\, e^{-j(2\pi/N)kn}$$

$$x[n] = \sum_{K=0}^{N-1} c_K e^{j(2\pi/N)kn}$$

- Aperiodic → Continuous, Periodic — Fourier Transform

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(\omega)\, e^{-j\omega n}$$

Properties of FT for Discrete signal:

1. Direct Transform : Take signal $x[n]$ & break it down to frequencies, $X(\omega)$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

2. Inverse Transform: Take frequencies and build back up to get signal $x[n]$

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(\omega)\, e^{j\omega n}\,d\omega$$

$$x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(\omega)$$

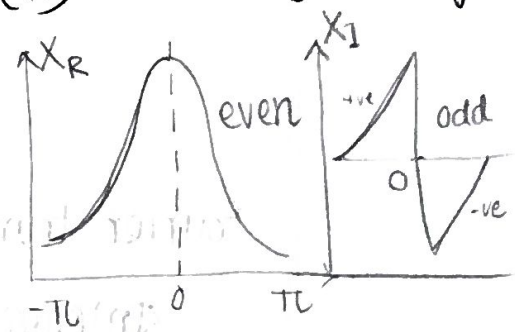
3. Symmetry (Real signals):

If signal is real, spectrum has mirror symmetry which reduces computation

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \underbrace{\sum x[n]\cos(\omega n)}_{X_R(\omega)} - \underbrace{j\sum x[n]\sin(\omega n)}_{X_I(\omega)}$$

For $-\omega$, $\cos(-\omega n) = \cos(\omega n) \rightarrow$ even
$\sin(-\omega n) = -\sin(\omega n) \rightarrow$ odd
} Hermitian Symmetry

$\therefore X_R(-\omega) = X_R(\omega) \rightsquigarrow$ even symmetry $\quad X_I(-\omega) = -X_I(\omega) \rightarrow$ odd symmetry



$\therefore$ Magnitude has even symmetry $|X(\omega)| = |X(-\omega)|$

$\therefore$ Phase has odd symmetry, $\angle X(-\omega) = -\angle X(\omega)$

4. Periodicity:

DTFT is periodic with a period $2\pi \rightarrow X(\omega + 2\pi) = X(\omega)$

5. Linearity:

If $\left.\begin{array}{l} x_1[n] \xleftrightarrow{\mathcal{F}} X_1(\omega) \\ x_2[n] \xleftrightarrow{\mathcal{F}} X_2(\omega) \end{array}\right\} a_1x_1[n] + a_2x_2[n] \xleftrightarrow{\mathcal{F}} a_1X_1(\omega) + a_2X_2(\omega)$

6. Time Shifting:

If $x[n] \longleftrightarrow X(\omega)$, then $X[n-k] = \sum_{n=-\infty}^{\infty} x[n-k]\, e^{-j\omega n} = \sum_{m=-\infty}^{\infty} x[m]\, e^{-j\omega(m+k)}$

$= e^{-j\omega k} \sum x[m] e^{-j\omega m} = e^{-j\omega k} \cdot X(\omega)$

$\therefore x[n-k] \longleftrightarrow e^{-j\omega k} X(\omega)$

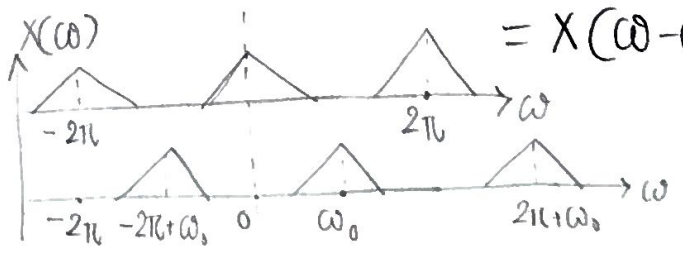
A shift in time-domain by k-samples, affects only phase of signal.

7. Frequency Shifting:

If $x[n] \leftrightarrow X(\omega)$ then $e^{j\omega_0 n} x[n] \Rightarrow \sum_{n=-\infty}^{\infty} (e^{j\omega_0 n} x[n]) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x[n]\, e^{-jn(\omega-\omega_0)}$

$= X(\omega - \omega_0)$

$\therefore e^{j\omega_0 n} x[n] \xleftrightarrow{\mathcal{F}} X(\omega - \omega_0)$



8 Convolution

$$x_1[n] * x_2[n] \xleftrightarrow{F} X_1(\omega) X_2(\omega)$$

→ convolution in time dom ⟷ multiplication in frequency dom

9. Modulation

$$x[n]\cos(\omega_0 n) \xleftrightarrow{F} \frac{1}{2}\left[X(\omega-\omega_0) + X(\omega+\omega_0)\right]$$

→ $$x[n]\cos(\omega_0 n) = x[n] \cdot \left[\frac{e^{j\omega_0 n}}{2} + \frac{e^{-j\omega_0 n}}{2}\right]$$

$$= \frac{1}{2}x[n]e^{j\omega_0 n} + \frac{1}{2}x[n]e^{-j\omega_0 n}$$

Fourier transform,

$$x[n]\cos(\omega_0 n) \longleftrightarrow \frac{1}{2}X(\omega-\omega_0) + \frac{1}{2}X(\omega+\omega_0)$$

10. Correlation

$$r_{x_1x_2}[m] \xleftrightarrow{F} S_{x_1x_2}(\omega) = X_1(\omega)X_2(-\omega)$$

Because,

$$r_{x_1x_2}[m] = \sum_{n=-\infty}^{\infty} x_1[n]\, x_2[n-m]$$

$$\therefore S_{x_1x_2}[\omega] = \sum_{m=-\infty}^{\infty}\left[\sum_{n=-\infty}^{\infty} x_1[n]\cdot x_2[n-m]\right]e^{-j\omega m}$$

let $k = n-m$, $m = n-k$, $e^{-j\omega(n-k)} = e^{-j\omega n}\cdot e^{j\omega k}$

$$\therefore S[\omega] = \sum_{n=-\infty}^{\infty} x_1[n]e^{-j\omega n} \cdot \sum_{k=-\infty}^{\infty} x_2[k]e^{j\omega k}$$

$$= X_1(\omega)\cdot X_2(-\omega)$$